

A three-axis superconducting guide-field and recoil detector system for absolute hadron beam polarimetry at the EIC using HJET

Design maturity and performance considerations

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Abstract

We present the conceptual design and performance evaluation of a compact, switchable three-dimensional magnetic holding-field system for the Hydrogen Jet (HJET) polarimeter at the Electron-Ion Collider (EIC). The system is designed to provide stable and well-defined guide fields for high-precision absolute proton beam polarimetry with a target precision of $\Delta P/P \approx 1\%$, while remaining compatible with the tight geometric constraints imposed by recoil-proton detection in the Coulomb-Nuclear Interference (CNI) region.

The proposed configuration employs three orthogonal pairs of superconducting coils generating the field components $\pm B_x$, $\pm B_y$, and $\pm B_z$, with only one component energized at a time. Detailed three-dimensional magnetic field calculations demonstrate that holding fields of up to 400 mT can be achieved in the target region with sufficient uniformity and manageable stray fields. At this field strength, beam-induced hyperfine transitions in the polarized hydrogen target are suppressed, effectively eliminating target depolarization and enabling beam polarization determination with controlled systematic uncertainties based on the measured target polarization. Numerical tracking studies quantify the impact of the guide field on recoil-proton trajectories and detector acceptance.

A dedicated Monte Carlo study is used to evaluate recoil-bunch assignment based on time-of-flight and recoil-energy consistency under realistic EIC timing conditions. The results demonstrate that reliable bunch identification can be achieved over the full CNI kinematic range for detector timing resolutions at the level of a few nanoseconds.

The compact geometry enables a staged, cryogen-free superconducting implementation with reduced technical complexity and cost compared to large-aperture magnet systems, providing a viable and scalable solution for high-precision HJET absolute beam polarimetry at the EIC.

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1 Introduction and motivation

The polarized atomic hydrogen gas jet (HJET) polarimeter is proposed to determine the absolute polarization of stored proton beams through elastic proton-proton scattering, with a targeted systematic uncertainty of $\Delta P/P \approx 1\%$, in the hadron storage ring of the Electron-Ion Collider (EIC). A critical element of the HJET system is the magnetic holding field, which preserves the polarization state of the hydrogen target atoms during beam-target interactions. Compared to operation at RHIC, the substantially higher bunch repetition frequency and shorter bunch length foreseen at the EIC shift the beam-induced electromagnetic field spectrum toward higher frequencies. These frequency components can resonantly drive hyperfine transitions in atomic hydrogen, leading to significant target depolarization if the holding field is insufficient. Detailed theoretical analysis of beam-induced depolarization effects [1] shows that operation at the RHIC holding-field strength of $B_0 = 120$ mT is not viable under EIC beam conditions. Instead, a substantially stronger holding field of at least $|\mathbf{B}| \geq 400$ mT is required to suppress resonant hyperfine transitions and ensure stable target polarization during EIC operation.

Beyond mitigating beam-induced depolarization, a three-axis, switchable superconducting holding-field system provides essential functionality for the EIC polarimetry program. Independent generation of holding fields along the three orthogonal directions (B_x , B_y , and B_z), with controlled polarity reversal for each axis, potentially enables access to all three components of the stored beam polarization vector, P_x , P_y , and P_z , at the interaction point. In particular, measurements with transverse holding-field orientations provide sensitivity to the transverse polarization components (P_x and P_y) through single-spin analyzing powers in elastic proton-proton scattering, while access to the longitudinal component (P_z) generally requires additional double-spin observables and corresponding experimental sensitivity. Field reversal (for example, switching between $+B_y$ and $-B_y$) allows systematic separation of spin-dependent scattering asymmetries from detector acceptance effects, non-flipping polarization components, and residual instrumental asymmetries, thereby significantly reducing systematic uncertainties in the extracted beam polarization.

The proposed detector concept employs a compact geometry with six silicon detector modules arranged symmetrically around the interaction region at a radial distance of approximately 220 mm. In contrast to the two-detector configuration at 750 mm used in the RHIC HJET system, the new layout significantly increases the effective acceptance for recoil protons in the kinematic region relevant for polarimetry. Since the elastic proton-proton analyzing power is sizable only in the Coulomb-nuclear interference region, corresponding to momentum transfers $|t| \sim 10^{-3} - 2 \times 10^{-2}$ GeV² [2], the useful acceptance is intrinsically restricted to recoil angles close to 90° with respect to the beam direction. Consequently, the dominant acceptance gain of the proposed detector geometry arises from the substantially increased azimuthal coverage rather than from an extension in polar angle. This enhanced azimuthal acceptance improves the statistical precision of polarization measurements and provides the flexibility required to accommodate frequent holding-field orientation changes without compromising overall measurement efficiency. A quantitative discussion of the acceptance gain, including the relevant kinematic constraints, is presented in Sec. 2.

A staged deployment strategy allows costs to be managed while establishing a clear upgrade path toward full three-axis holding-field capability. In Phase 1, only the B_y holding field will be implemented, with the detector system configured in the horizontal ring plane. The initial detector complement will comprise two silicon detector modules, compared to the full six-detector configuration foreseen for the complete system, thereby limiting the initial scope and cost of the installation. This baseline configuration satisfies the immediate operational requirements for absolute beam polarimetry and establishes the core HJET capability for the EIC. Despite its reduced detector complement, the Phase 1 layout provides a substantial increase in effective acceptance relative to the RHIC HJET system, primarily through improved azimuthal coverage in the kinematic region relevant for polarimetry (see Sec. Sec. 2).

Critically, the vacuum chamber, cryogenic infrastructure, and most of the essential services required for superconducting magnet operation will be installed already during Phase 1. This ensures that the initial implementation does not constrain future upgrades. In Phase 2, to be realized when additional resources become available, the B_x and B_z field-generation hardware will be added to enable full three-axis holding-field operation with controlled field reversal. This staged approach distributes the capital expenditure across multiple funding cycles while preserving the flexibility required for a complete and fully functional three-axis polarimeter system.

The technical requirements for the magnetic holding field system are summarized in Table 1. The specified central field magnitude, field uniformity¹, and switching time are motivated by the need to preserve the polarization of the atomic hydrogen target, suppress beam-induced depolarization mechanisms, minimize spin-state mixing systematics, and maintain an acceptable data-taking duty factor. Long-term reliability over decades-scale operation is an essential requirement, given the anticipated lifetime of the EIC experimental program and the associated infrastructure investment.

Table 1: Magnetic holding field system requirements.

Parameter	Specification	Notes
Central field magnitude	$ \mathbf{B} \geq 400 \text{ mT}$	Suppress beam-induced depolarization of the hydrogen target.
Field uniformity	$\Delta B/B \sim 1 \times 10^{-3}$	Order-of-magnitude target in the central interaction volume (about 10 mm^3), subject to later refinement, as nuclear target polarization depends non-linearly on $ \mathbf{B} $.
Operational modes	$\pm B_x$, $\pm B_y$, or $\pm B_z$	Baseline: one axis energized at a time. Upgrade: general orientations via superposition of fields.
Field switching time	Order 1 s to 10 s	Typical time scale for holding-field reorientation, chosen to limit duty-factor loss.
Dwell time per field orientation	$\sim 100 \text{ s}$	Representative integration time per field setting for required statistical precision.
Operational lifetime	Decades	Maintainable design for the full EIC program lifetime.

¹Table 1 summarizes order-of-magnitude target values used to guide the initial design space, while Table 4 lists realistic field uniformities obtained from numerical field calculations for the present geometry.

2 Boundary conditions and hard constraints

2.1 Geometry and detector acceptance

The magnetic guide-field system must be integrated into a cylindrical vacuum chamber with its symmetry axis aligned along the stored hadron beam direction, defined as the z axis. The system operates in ultrahigh vacuum shared with the circulating beam and the polarized hydrogen jet target. All magnet components, including superconducting coils, thermal management structures, and current leads, must therefore be fully vacuum compatible and constructed from low-outgassing materials suitable for long-term UHV operation.

The central beam aperture maintains a clear bore of at least 100 mm diameter along the z axis and remains entirely free of magnetic material and structural elements, in order to preserve beam quality and minimize perturbations to the circulating hadron beam². A conservative clearance criterion of at least 10σ should be applied at the HJET location, using the largest normalized horizontal gold emittance listed in Table 1.2 of the EIC Conceptual Design Report [4] and assuming equal horizontal and vertical emittances at injection.

The interaction region is centered at $z = 0$ and must remain free of magnet structures in order to preserve full azimuthal acceptance for recoil-proton detection. An RF shield will be incorporated in a later design iteration to suppress beam-induced electromagnetic fields.

The detector configuration employs six silicon detector modules arranged in three opposing pairs at a radial distance of $r \approx 220$ mm from the interaction point. Each detector module consists of two closely spaced silicon detector layers forming a compact two-layer unit, as illustrated in Fig. 1. The three detector pairs are located along the horizontal (x) axis and along the $\pm 45^\circ$ diagonals in the transverse plane, as illustrated in Fig. 1. In the staged deployment approach, Phase 1 implements only the horizontal detector pair together with the B_y holding-field coils, establishing the baseline polarimetry capability at reduced cost. Phase 2 adds the diagonal detector pairs as well as the B_x and B_z coil systems, enabling full azimuthal coverage and complete three-axis holding-field capability.

Each detector has an active area of approximately $100 \text{ mm} \times 100 \text{ mm}$. The effective acceptance is defined by straight-line trajectories from the target center to the detector edges, which determine the angular coverage relevant for recoil-proton detection. This acceptance geometry directly constrains the placement of the magnet coils, since current-carrying conductor legs must be positioned such that they do not intrude into or shadow the detector acceptance cones. As discussed in Sec. 4 and summarized in Table 2, this requirement places the transverse coil legs at ± 33.1 mm, aligned with the $\pm 22.5^\circ$ rays from the interaction point.

2.2 Acceptance considerations and kinematic constraints

The compact detector geometry provides a substantial increase in geometric solid-angle coverage compared to the present RHIC HJET configuration. For a flat detector at distance r with active area A , the geometric solid angle scales approximately as $\Omega \approx A/r^2$. However, the physically relevant acceptance for HJET polarimetry is constrained to recoil angles close to 90° with respect to the beam direction, because the elastic pp analyzing power is sizable only in the Coulomb-nuclear interference region, $|t| \sim 10^{-3} - 2 \times 10^{-2} \text{ GeV}^2$ [2]. In this regime, the useful gain of a compact detector layout is therefore governed primarily by azimuthal coverage rather than by an extension in polar angle.

²The assumed clear bore is expected to be sufficient, but must be verified against the maximum beam envelope at injection. The limiting case is the heavy-ion (gold) beam, for which a conservative clearance of at least 10σ is recommended [3].

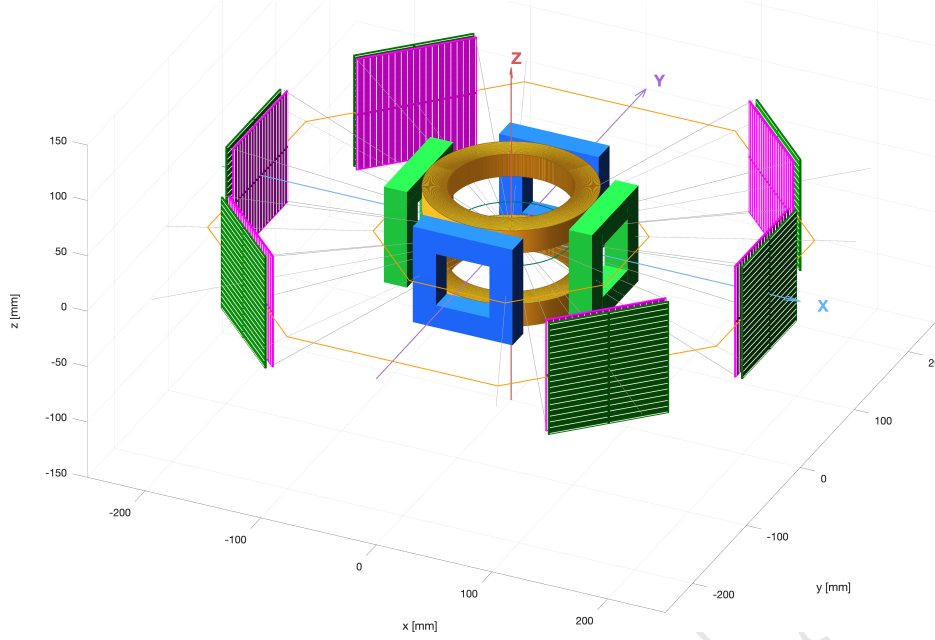


Figure 1: Three-dimensional (x, y, z) view of the compact magnet and detector geometry at the interaction point ($z = 0$). The ion beam propagates along the $+z$ direction, the atomic hydrogen beam is oriented along the $-y$ direction (y is perpendicular to the ring plane), and the x axis points radially inward toward the center of the ring. Twelve silicon detector layers, arranged as six detector modules, are positioned at $r = 220$ mm in three opposing pairs along the horizontal axis and the $\pm 45^\circ$ diagonals. Each module consists of two closely spaced single-sided silicon strip detector layers. The first layer, shown in magenta, is oriented to provide azimuthal angle information, while the second layer, shown in dark green, is oriented perpendicular to the first layer to provide an independent spatial coordinate. The strip orientations are indicated schematically by representative strip lines on the detector layers with only a limited number of strips shown for illustrative purposes. Each sensor is segmented into two parallel strip columns, as specified in Table 9. Straight lines indicate acceptance boundaries from the target center to the detector edges. Superconducting coil cross sections for B_x and B_y generation are shown in blue and green, positioned at $x, y = \pm 80$ mm and aligned with the $\pm 22.5^\circ$ rays to avoid acceptance shadowing. The circular B_z coil pair, concentric with the beam axis and located at $|z| = \pm 33.1$ mm, is shown in orange.

For a detector of active width w at radial distance r from the interaction point, the azimuthal angular coverage is, to good approximation,

$$\Delta\varphi \simeq 2 \arctan\left(\frac{w/2}{r}\right). \quad (1)$$

Using $w = 100$ mm yields $\Delta\varphi_{\text{RHIC}} \simeq 2 \arctan(50 \text{ mm}/750 \text{ mm}) \approx 7.63^\circ$ for the RHIC layout ($r \simeq 750$ mm), and $\Delta\varphi_{\text{EIC}} \simeq 2 \arctan(50 \text{ mm}/220 \text{ mm}) \approx 25.8^\circ$ for the proposed EIC modules ($r = 220$ mm).

2.2.1 Effective sensitivity for vertical polarization measurements

For a vertical beam polarization component P_y (with the hadron beam along $\vec{e}_z$), the single-spin azimuthal modulation of the elastic yield can be written schematically as $N(\varphi) \propto 1 + A_N P_y \cos \varphi$

Table 2: Coil geometry parameters for the transverse (B_x , B_y) and longitudinal (B_z) holding-field coils.

Parameter	Value
Transverse coil center position	$x = \pm 80$ mm (B_x), $y = \pm 80$ mm (B_y)
Transverse leg positions	± 33.1 mm along orthogonal directions
Leg spacing (center-center)	66.2 mm
Conductor cross section (all coils)	20 mm $\times$ 20 mm
Transverse coil outer side length	86.2 mm
Mean current perimeter (B_x , B_y)	265 mm
B_z coil mean radius	60 mm
B_z coil axial positions	$z = \pm 33.1$ mm
B_z coil inner / outer radius	50 mm / 70 mm

(up to a choice of azimuth convention). Consequently, the statistical sensitivity to P_y from a detector at azimuth φ_i scales with $\cos \varphi_i$, and the corresponding contribution to the inverse variance scales with $\cos^2 \varphi_i$. For equal detector areas and comparable efficiencies, a convenient geometric proxy for the effective figure-of-merit is therefore

$$F_y \propto \sum_{i=1}^N \Delta\varphi_i \cos^2 \varphi_i \simeq \Delta\varphi \sum_{i=1}^N \cos^2 \varphi_i, \quad (2)$$

where $\Delta\varphi$ is given by Eq. (1). Since the statistical uncertainty scales as $\sigma(P_y) \propto 1/\sqrt{F_y}$, the time required to reach a fixed statistical precision scales as $t \propto 1/F_y$.

For the RHIC reference and EIC Phase 1 layouts (two detectors on the horizontal axis, $\varphi = 0^\circ, 180^\circ$), one has $\cos^2 \varphi = 1$ for both modules. For the full six-detector EIC Phase 2 layout ($\varphi = 0^\circ, 180^\circ, \pm 45^\circ, \pm 135^\circ$), four modules contribute with $\cos^2 \varphi = 1/2$ and two with $\cos^2 \varphi = 1$, giving $\sum \cos^2 \varphi = 4$.

The resulting geometric azimuthal coverages and the corresponding P_y -weighted effective coverages are summarized in Table 3. The last column lists the relative data-taking time required to reach the same statistical uncertainty on P_y , normalized to the RHIC reference case.

Table 3: Azimuthal acceptance comparison for recoil-proton detection in the CNI region. The per-module azimuthal coverage is $\Delta\varphi$ from Eq. (1). The effective P_y figure-of-merit proxy uses the weighting in Eq. (2), i.e. $\sum_i \Delta\varphi_i \cos^2 \varphi_i$. Relative time is normalized to RHIC assuming $t \propto 1/F_y$.

Configuration	r [mm]	w [mm]	N	$\Delta\varphi$ [$^\circ$]	$N\Delta\varphi$ [$^\circ$]	F_y proxy [$^\circ$]	t/t_{RHIC}
RHIC (2 det., horiz.)	750	100	2	7.63	15.3	15.3	1.00
EIC Phase 1 (2 det., horiz.)	220	100	2	25.8	51.6	51.6	0.296
EIC Phase 2 (6 det.)	220	100	6	25.8	155	103	0.149

After proper azimuthal projection of the recoil-proton asymmetries, the effective statistical gain of the proposed EIC detector system is reduced relative to the purely geometric estimate, but still amounts to approximately a factor of 6–7 compared to the present RHIC HJET configuration.

The determination of P_x can be addressed in an analogous manner using the corresponding

transverse analyzing power, whereas access to the longitudinal component P_z is more involved, as it relies on double-spin observables that are presently less well constrained experimentally.

The gain factor quoted above refers to the reduction in measurement time at fixed statistical precision. In addition, the EIC proton beam intensity is expected to be higher by roughly a factor of three compared to RHIC, which directly translates into a further reduction of the required measurement time. Taken together, for the proposed full system with six detectors, an overall time reduction by approximately a factor of $6.8 \times 3 \approx 20$ can therefore be expected relative to the present RHIC HJET configuration.

2.3 Operational sequencing and downtime budget

The baseline operational cycle sequences through the three orthogonal holding-field orientations, with dwell times of order 100 s per orientation to accumulate sufficient statistics and field switching times of order 1 s to 10 s between orientations. For a complete three-axis cycle with dwell time $t_{\text{dwell}} \approx 100$ s and representative switching time $t_{\text{switch}} \approx 5$ s, the fractional downtime associated with field reorientation is

$$f_{\text{down}} \approx \frac{3 t_{\text{switch}}}{3 t_{\text{dwell}} + 3 t_{\text{switch}}} \approx \frac{15 \text{ s}}{315 \text{ s}} \approx 4.8 \%. \quad (3)$$

This estimate assumes three switching intervals per full cycle (one between successive field orientations) and negligible additional settling time once the target field is reached. The achievable duty factor will ultimately depend on the realizable ramp rates, current stability, and reproducibility of the superconducting magnet system, which will be established through field measurements and operational experience. Superconducting magnets offer the advantage of programmable current ramps and precise field control, allowing the switching sequence to be optimized to minimize systematic effects while maintaining a high data-taking efficiency.

2.4 Stray-field constraint at the quadrupole mass analyzer

A quadrupole mass analyzer (QMA) used for residual gas analysis and target beam diagnostics will be located approximately 350 mm from the interaction region in the vertical direction (below the target chamber). The QMA must be capable of scanning in the xz plane over a range of approximately ± 40 mm around the y axis to enable profile measurements of the atomic hydrogen beam composition. These profile measurements are essential for determining the unpolarized H_2 fraction in the atomic beam, which directly affects systematic uncertainties in the target polarization determination.

Quadrupole mass spectrometers are sensitive to external magnetic fields, which can distort ion trajectories and degrade mass resolution and transmission efficiency. Conservative practice in accelerator and vacuum instrumentation targets stray field levels at or below 0.1 mT at the QMA location, though vendor-specific tolerances and local magnetic shielding options must be evaluated [5, 6]. The scanning requirement adds complexity: the stray field must remain below tolerance not only at the nominal QMA position but throughout the entire scanning range, and any magnetic shielding solution must accommodate the mechanical motion without interference.

The scanning operation with the QMA will be performed only intermittently and does not coincide with polarized-beam data taking. During QMA scans, the superconducting magnetic guide field will be switched off, eliminating stray magnetic fields at the analyzer location. Under these conditions, no additional stray-field mitigation measures are required for QMA operation.

2.5 Field polarity reversal and polarization separation capability

The magnetic holding-field system must provide full polarity reversal of all three field components, $\pm B_x$, $\pm B_y$, and $\pm B_z$, as well as the capability to switch the guide field completely off. This functionality is essential for controlling systematic effects in absolute polarimetry and for separating flipping and non-flipping contributions to the measured target and beam polarization [7].

The availability of both field polarities enables an unambiguous and separate determination of the polarization associated with the injected hydrogen hyperfine state combinations $|1\rangle + |4\rangle$ and $|2\rangle + |3\rangle$. Reversal of the magnetic guide-field direction reverses the sign of the target polarization component that follows the field, while non-flipping contributions remain unchanged. By comparing measurements taken with opposite field polarities, the polarization contributions of the two injected hyperfine-state combinations can therefore be disentangled experimentally.

In addition, field-reversal symmetries provide powerful internal consistency checks on the polarization analysis and sensitivity to residual systematic effects. The ability to perform controlled field reversals thus constitutes a hard boundary condition for the magnetic guide-field system design.

3 Magnet System Design

3.1 Design overview and field generation concept

The holding field system employs three pairs of orthogonal superconducting coils to generate switchable magnetic fields B_x , B_y , and B_z of up to 400 mT in the central interaction region. Each field component is produced by energizing one coil pair while the other two pairs remain de-energized, enabling rapid switching between field orientations without mechanical motion. The transverse field components B_x and B_y are generated by square-coil pairs positioned at $x = \pm 80$ mm and $y = \pm 80$ mm, respectively, with conductor legs aligned to preserve detector acceptance as detailed below and illustrated in Fig. 1. The longitudinal field component B_z is generated by a circular superconducting coil pair concentric with the beam axis and positioned symmetrically upstream and downstream of the interaction region.

The compact geometry preserves full detector acceptance while maintaining a central material-free volume of $100 \text{ mm} \times 100 \text{ mm} \times 100 \text{ mm}$ around the interaction point. All coils operate within the main vacuum chamber and employ dry-wound construction with vacuum-compatible insulation materials to minimize outgassing. The modest stored energy per coil pair (few kilojoules) simplifies quench protection and enables fast field ramping for operational flexibility.

3.2 Geometric layout and detector acceptance constraint

The transverse coil geometry is defined in the xy plane, as illustrated in Fig. 1. The B_y holding field is generated by a pair of square superconducting coils centered at $y = \pm 80$ mm, while the B_x coils are obtained by a 90° rotation about the z axis and are centered at $x = \pm 80$ mm. Each coil consists of four current-carrying legs forming a square loop in a plane perpendicular to the beam axis, with the conductor legs running parallel to z through the interaction region.

A critical geometric constraint arises from the requirement to preserve full azimuthal detector acceptance. The six silicon detector modules are positioned at a radial distance of $r = 220$ mm from the interaction point and define straight-line acceptance boundaries from the target center to the detector edges, as indicated by dashed lines in Fig. 1. The coil conductor legs must be placed such that they do not intersect or shadow these acceptance cones.

This constraint is satisfied by aligning the transverse coil legs with the $\pm 22.5^\circ$ rays from the interaction point. For a coil centered at a transverse distance $d = 80$ mm from the origin, the

required leg positions follow directly from geometry,

$$x_{\text{leg}} = d \tan(22.5^\circ) \approx 33.1 \text{ mm} . \quad (4)$$

Accordingly, the four legs of a B_x coil are located at $y, z = \pm 33.1 \text{ mm}$, while those of a B_y coil are located at $x, z = \pm 33.1 \text{ mm}$. This results in a square current loop with a center-to-center leg spacing of 66.2 mm. The corresponding mean current perimeter per turn is

$$\ell_{\text{turn}} = 4 \times 66.2 \text{ mm} = 265 \text{ mm} . \quad (5)$$

The complete geometric parameters for the transverse (B_x, B_y) and longitudinal (B_z) holding-field coils are summarized in Table 2.

3.3 Conductor cross-section and current density

Each coil leg occupies a square cross section of $20 \text{ mm} \times 20 \text{ mm}$, corresponding to a total conductor pack area of 400 mm^2 . For first-order design estimates, an effective superconducting current-carrying area of $10 \text{ mm} \times 10 \text{ mm} = 100 \text{ mm}^2$ is assumed. The remaining volume is allocated to electrical insulation, stabilizer material (copper or aluminum), mechanical reinforcement, and thermal conduction paths to the cryocooler cold head.

This conservative ratio of total conductor pack area to superconducting area of approximately 4:1 provides ample margin for mechanical stability, quench protection, and thermal management in an ultrahigh-vacuum environment. It is adequate for the present conceptual design and will be refined once a specific superconducting conductor and winding technology are selected in consultation with vendors.

3.4 Magnetic field calculation and required ampere-turns

The magnetic field generated by the square coils was calculated numerically using the Biot-Savart law,

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{I d\boldsymbol{\ell} \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} , \quad (6)$$

by discretizing each of the four straight leg segments into small line elements and summing their contributions [8]. The calculation treats the current as flowing along the conductor centerline (idealized line-current approximation), which is adequate for determining the total ampere-turn requirement and the field magnitude at the interaction point.

Field uniformity studies accounting for the finite $20 \text{ mm} \times 20 \text{ mm}$ conductor cross-section and the full three-dimensional coil geometry will be performed using magnetostatic finite-element simulations as part of the detailed engineering design phase.

3.4.1 Analytical field estimate for a square loop

For reference, the magnetic field of a square current loop can be calculated analytically using the Biot-Savart law. The complete derivation is provided in Appendix A. The result for the field at the center of a square loop of side length L carrying current I is

$$B_{\square}(0) = \frac{2\sqrt{2}\mu_0 I}{\pi L} . \quad (7)$$

For a field point located along the axis perpendicular to the loop plane at distance z from the center, the field is given by

$$B_{\square}(z) = \frac{4\mu_0 I}{\pi L} \frac{1}{[1 + (2z/L)^2] \sqrt{2 + (2z/L)^2}}. \quad (8)$$

For the present geometry with $L = 66.2$ mm, a single turn with $I = 1$ A at the loop center produces

$$B_{\square}(0) = \frac{2\sqrt{2}\mu_0}{\pi L} \approx 1.71 \times 10^{-5} \text{ T}. \quad (9)$$

For a coil pair with loop planes at $x = \pm d$ (or $y = \pm d$) with $d = 80$ mm, each loop is at axial distance $z = d$ from the interaction point. With $2z/L = 160/66.2 = 2.417$, Eq. (8) yields a single-loop contribution at the origin of

$$B_{\text{single loop}}(z = d) = \frac{4\mu_0}{\pi L} \frac{1}{[1 + (2d/L)^2] \sqrt{2 + (2d/L)^2}} \approx 1.26 \times 10^{-6} \text{ T per A-turn}. \quad (10)$$

Both loops contribute in the same direction at the origin, so

$$B_{\text{pair, thin wire}}(0) \approx 2.52 \times 10^{-6} \text{ T per A-turn}. \quad (11)$$

To reach the design field of 400 mT, the required total ampere-turns for one transverse coil pair are

$$NI \approx \frac{0.400 \text{ T}}{2.52 \times 10^{-6} \text{ T}} \approx 1.59 \times 10^5 \text{ A-turn}. \quad (12)$$

A baseline of $N = 100$ turns per coil is adopted, yielding an operating current of

$$I \approx 1.59 \text{ kA}. \quad (13)$$

With an effective current-carrying area of 100 mm^2 , the resulting current density is

$$J \approx 15.9 \text{ A/mm}^2, \quad (14)$$

while the engineering current density referenced to the full $20 \text{ mm} \times 20 \text{ mm}$ conductor pack is

$$J_{\text{eng}} \approx 4.0 \text{ A/mm}^2. \quad (15)$$

3.5 Conductor length, inductance, and stored energy

The total conductor length per coil is

$$L_{\text{coil}} = N \times \ell_{\text{turn}} = 100 \times 265 \text{ mm} \approx 26.5 \text{ m}, \quad (16)$$

corresponding to approximately 53 m for a complete coil pair.

The inductance L of a coil pair is determined by its geometry and can be inferred from the total magnetic energy stored in the generated field according to

$$E_{\text{stored}} = \frac{1}{2} L I^2. \quad (17)$$

At the conceptual design level, an order-of-magnitude estimate of the stored energy can be obtained from the magnetic energy density,

$$E_{\text{stored}} \sim \frac{1}{2\mu_0} \int B^2(\mathbf{r}) dV \approx \frac{B^2 V}{2\mu_0}, \quad (18)$$

where V denotes a representative central volume over which the field is of order B .

For the present design, taking $B = 400$ mT over a characteristic volume of order $(100 \text{ mm})^3$ yields an inductance estimate of order $L \sim 3$ mH to 6 mH. Using the nominal operating current inferred from the numerical Biot-Savart scaling, $I \approx 1.6$ kA (for $N = 100$ turns per coil, corresponding to $NI \approx 1.6 \times 10^5$ A-turn for one coil pair), the corresponding stored magnetic energy is

$$E_{\text{stored}} \approx \frac{1}{2} LI^2 \approx \frac{1}{2} \times 4.5 \text{ mH} \times (1.6 \text{ kA})^2 \approx 5.7 \text{ kJ}. \quad (19)$$

This modest stored energy significantly simplifies quench protection and energy extraction. A precise determination of L and E_{stored} will be obtained in a later design phase from three-dimensional field simulations by direct numerical integration of $B^2(\mathbf{r})$ over space.

3.6 Field ramping and switching

The operational cycle requires switching between field orientations on timescales. For a representative ramp time of 5 s, the required ramp rate is

$$\frac{dI}{dt} \approx \frac{1.59 \text{ kA}}{5 \text{ s}} \approx 317 \text{ A/s}. \quad (20)$$

The induced voltage across the coil pair during ramping is

$$V = L \frac{dI}{dt} \approx 4.5 \text{ mH} \times 317 \text{ A/s} \approx 1.4 \text{ V}. \quad (21)$$

This voltage level is modest and well within the capability of standard bipolar power supplies for superconducting magnets. The power dissipation associated with ramping is negligible compared to the steady-state cryogenic heat load, and the ramping overhead contributes only a few percent to the total operational cycle, as quantified in Sec. 2.3.

3.7 Numerical field calculations and validation

To validate the analytical estimates and to assess field uniformity in the interaction region, three-dimensional magnetic field calculations were performed using a numerical implementation of the Biot-Savart law. The full coil geometries described in Table 2 were modeled explicitly, including the exact coil positions, orientations, and current paths. The calculations were carried out in MATLAB by discretizing each current-carrying conductor into straight line segments and summing their individual field contributions.

For each coil pair (B_x , B_y , and B_z), the current was scaled such that the target field magnitude of $|\mathbf{B}| = 400$ mT was obtained at the interaction point. Figure 2, Fig. 3, and Fig. 4 show the resulting field components along the principal coordinate axes through the origin. In all cases, the numerical results confirm the analytical ampere-turn estimates derived in Sec. 3.4.1 to within a few percent.

The calculated field profiles demonstrate that the magnetic field in the central interaction region is dominated by the intended holding-field component defined by the selected coil pair, while transverse components remain small. For each configuration, we define the longitudinal field component as

$$B_{\parallel}(\vec{r}) = \vec{B}(\vec{r}) \cdot \vec{e}_i, \quad (22)$$

with $\vec{e}_i \in \{\vec{e}_x, \vec{e}_y, \vec{e}_z\}$ corresponding to the B_x , B_y , and B_z coil pairs. Quantitative measures of field alignment and uniformity within the central interaction volume of approximately $10 \text{ mm} \times 10 \text{ mm} \times 10 \text{ mm}$ are summarized in Table 4. In this volume, the mean field magnitude satisfies $|\langle \vec{B} \rangle| \approx$

0.400 T for all three configurations. The rms relative variation of the longitudinal component, $\text{rms}(\Delta B_{\parallel}/B_{\parallel})$, is 8.669×10^{-3} for the B_x and B_y configurations and 8.340×10^{-4} for the B_z configuration. The transverse field content, expressed as $\text{rms}(|\vec{B}_{\perp}|/B_{\parallel})$, remains small, with values of 1.394×10^{-2} for B_x and B_y and 1.283×10^{-3} for B_z .

These results demonstrate that the present compact coil geometry satisfies the required field magnitude and uniformity specifications summarized in Table 1, while preserving full detector acceptance. If higher field uniformity were required in future operation, the coil geometry itself could be modified or optimized to further reduce spatial variations of $B_{\parallel}$ without changing the overall detector concept.

Table 4: Magnetic-field uniformity and alignment in the central interaction volume for the three coil-pair configurations. The metrics are evaluated within the cube $|x|, |y|, |z| \leq 5$ mm using the same three-dimensional field maps as in Figs. 2–4. The nominal component $B_{\parallel}$ is the field component parallel to the intended holding-field direction defined by the selected coil pair. Reported are the mean nominal field $\langle B_{\parallel} \rangle$, the relative rms variation $\sigma(B_{\parallel})/\langle B_{\parallel} \rangle$, the relative maximum deviation $\max |B_{\parallel} - \langle B_{\parallel} \rangle|/\langle B_{\parallel} \rangle$, and the rms transverse fraction $\text{rms}(B_{\perp})/\langle B_{\parallel} \rangle$ that quantifies field misalignment.

Coil pair	$\langle B_{\parallel} \rangle$ [T]	$\sigma(B_{\parallel})/\langle B_{\parallel} \rangle$	$\max B_{\parallel} - \langle B_{\parallel} \rangle /\langle B_{\parallel} \rangle$	$\text{rms}(B_{\perp})/\langle B_{\parallel} \rangle$
B_x pair	0.399968	8.669×10^{-3}	1.497×10^{-2}	1.394×10^{-2}
B_y pair	0.399968	8.669×10^{-3}	1.497×10^{-2}	1.394×10^{-2}
B_z pair	0.400028	8.340×10^{-4}	1.492×10^{-3}	1.283×10^{-3}

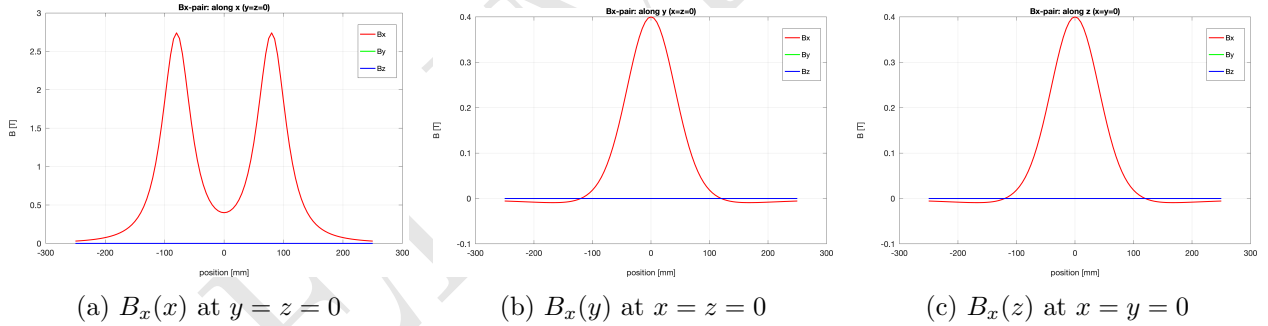


Figure 2: Numerically calculated magnetic field component B_x generated by the B_x coil pair. Shown are one-dimensional field profiles along the three principal axes through the interaction point, obtained from the Biot-Savart calculation described in Sec. 3.7.

3.8 Impact of the holding field on recoil proton hit positions

The strong transverse holding field required to suppress beam-induced depolarization inevitably leads to a magnetic deflection of recoil protons before they reach the silicon detector planes. A quantitative estimate of the resulting hit displacements is presented in Appendix B, where the proton trajectories are evaluated using the numerically calculated magnetic field of the holding-field coils.

For the baseline geometry and a holding field of 400 mT, the calculated transverse displacements at the detector surface remain at the level of a few centimeters even for the lowest momentum-

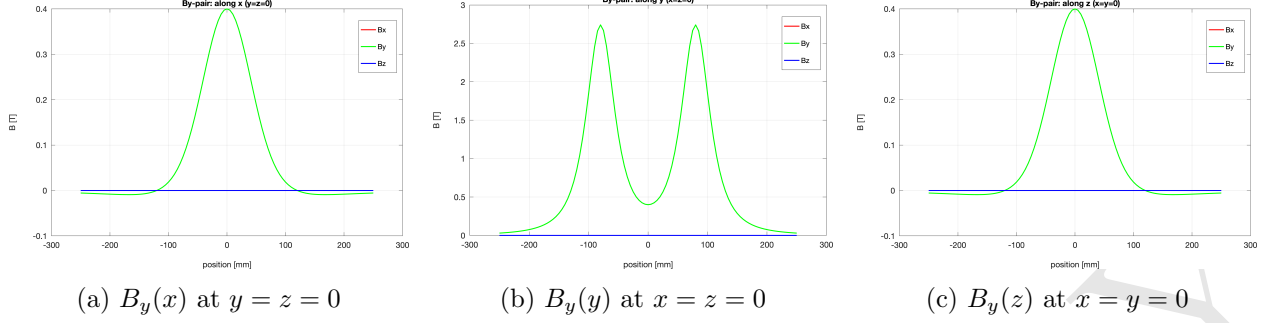


Figure 3: Numerically calculated magnetic field component B_y generated by the B_y coil pair. Shown are one-dimensional field profiles along the three principal axes through the interaction point, obtained from the Biot-Savart calculation described in Sec. 3.7.

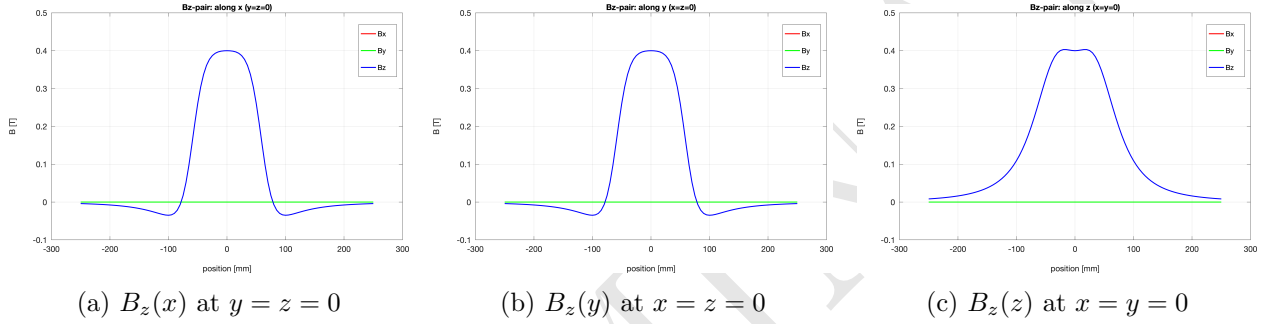


Figure 4: Numerically calculated magnetic field component B_z generated by the B_z coil pair. Shown are one-dimensional field profiles along the three principal axes through the interaction point, obtained from the Biot-Savart calculation described in Sec. 3.7.

transfer values in the CNI region. These shifts are well within the active detector area and do not compromise detector acceptance or tracking capability.

Importantly, the magnetic field does not affect the recoil kinetic energy T_R or the magnitude of the recoil momentum p_R , which are determined solely by the momentum transfer t . Since the polarimetry analysis is based on the reconstruction of t via T_R (or equivalently p_R), the holding field does not introduce a bias in the extracted analyzing power or polarization observables. The magnetic deflection only modifies the mapping between recoil momentum and hit position on the detector surface, an effect that can be fully accounted for in the detector geometry and reconstruction software.

3.9 Path forward

The chosen coil configuration satisfies the geometric and detector acceptance constraints while achieving the required 400 mT holding field. For the present geometry ($d = 80$ mm, $s = 33.1$ mm), the numerical Biot-Savart scaling yields a required ampere-turn level of $NI \approx 1.59 \times 10^5$ A per coil to reach $B(0) = 400$ mT; adopting a baseline of $N = 100$ turns therefore implies an operating current of $I \approx 1.59$ kA per coil. With an effective current-carrying area of 100 mm², the resulting current density is $J \approx 15.9$ A/mm², while the engineering current density referenced to the full 20 mm $\times$ 20 mm conductor pack is $J_{\text{eng}} \approx 4.0$ A/mm². These values remain within acceptable ranges for superconducting technologies but will require careful thermal management and

conductor stabilization. The system enables cost-effective implementation using either HTS or LTS superconducting technology, as detailed in Sec. 4. It is well suited for staged deployment, with Phase 1 implementing B_y generation only (two coils) and Phase 2 adding B_x and B_z capability (four additional coils) when funding permits. Next engineering tasks include three-dimensional field simulations to verify uniformity requirements and to refine the ampere-turn estimate accounting for finite conductor cross section, thermal and mechanical analysis of vacuum-compatible coil support structures, and vendor consultation to finalize conductor specifications and fabrication approaches.

4 Superconducting magnet approach

4.1 Technology options and cost comparison

The compact coil geometry developed in the preceding sections enables the use of superconducting technology at substantially reduced scale and cost compared to conventional large-bore magnet systems. Two primary superconducting options were evaluated: high-temperature superconductors (HTS) operating at liquid nitrogen temperature (77 K) and low-temperature superconductors (LTS) operating at liquid helium temperature (4.2 K). A permanent-magnet solution based on Halbach arrays was initially considered but not adopted, because such systems cannot be fully switched off. While Halbach magnets can be reoriented mechanically to change the field direction, a true zero-field state cannot be achieved. This is incompatible with operational requirements of the HJET system, in particular periodic quadrupole mass analyzer scans, which require the magnetic guide field to be completely disabled. Superconducting coils provide the required capability to ramp the field continuously from zero to the nominal operating value and back, with precise control over field magnitude and orientation, and are therefore the preferred solution for the present application.

The cost estimates discussed below are based on the single-coil geometry defined in Table 2, with $N = 100$ turns per coil and a mean perimeter of 265 mm per turn. The required operating current follows from the numerical Biot-Savart field calculations and amounts to approximately 1.6 kA per coil, corresponding to a total of 1.6×10^5 A-turns per coil pair to generate a holding field of 400 mT at the interaction point.

Table 5 summarizes the resulting cost estimates and key characteristics for the superconducting technology options considered.

Table 5: Superconducting technology options comparison for compact holding-field coils (geometry per Table 2, $N = 100$ turns/coil, mean perimeter 265 mm per turn, operating current $I \simeq 1.6$ kA per coil from the numerical Biot-Savart scaling to 400 mT at the interaction point).

Parameter	HTS @ 77 K (REBCO)	LTS @ 4.2 K (NbTi)	LTS @ 4.2 K (Nb ₃ Sn)
Conductor specification	2G REBCO tape; typically 6 to 12 tapes in parallel (tape-grade and margin dependent), $I_c \sim 200$ A to 400 A/tape @ 77 K	NbTi multifilament (Rutherford cable or stabilized wire), mature industrial supply	Nb ₃ Sn (react-and-wind or wind-and-react), brittle; fabrication more complex
Required current density (this design)	Effective superconductor area ≈ 100 mm ² and coil-pack area 400 mm ² : $J \approx 16$ A/mm ² (effective), $J_{\text{eng}} \approx 4$ A/mm ² (engineering).		
Cost breakdown (per coil):			
Conductor (Option B, scaled with tape count)	16 k\$ to 32 k\$	6 k\$ to 12 k\$	10 k\$ to 18 k\$
Coil winding & support structure	12 k\$ to 18 k\$	10 k\$ to 15 k\$	15 k\$ to 22 k\$
Cryostat & thermal mgmt	5 k\$ to 10 k\$	15 k\$ to 25 k\$	15 k\$ to 25 k\$
Current leads	5 k\$ to 8 k\$	5 k\$ to 8 k\$	5 k\$ to 8 k\$
<i>Total per coil</i>	38 k\$ to 68 k\$	36 k\$ to 60 k\$	45 k\$ to 73 k\$
Cooling system (one-time):			
Infrastructure	Closed-cycle cryocooler (77 K)	Closed-cycle cryocooler (4.2 K)	Closed-cycle cryocooler (4.2 K)
Cost	5 k\$ to 10 k\$	50 k\$ to 100 k\$	50 k\$ to 100 k\$
Typical cooldown time (RT $\rightarrow$ operating T)	~ 6 h to 12 h (to 77 K)	~ 24 h to 72 h (to 4.2 K)	~ 48 h to 96 h (to 4.2 K, strain-limited)
Power supply (per coil):	Bipolar DC, 1.6 kA, ± 15 V, 20 k\$ to 35 k\$		
Phase costs:			
Phase 1 (B_y , 2 coils + 1 PS)	96 k\$ to 171 k\$	92 k\$ to 155 k\$	110 k\$ to 181 k\$
Cooling infrastructure	5 k\$ to 10 k\$	50 k\$ to 100 k\$	50 k\$ to 100 k\$
Phase 1 subtotal	101 k\$ to 181 k\$	142 k\$ to 255 k\$	160 k\$ to 281 k\$
Phase 2 (add 4 coils + 2 PS)	192 k\$ to 342 k\$	184 k\$ to 310 k\$	220 k\$ to 365 k\$
Total system cost (Phase 1+2)	293 k\$ to 523 k\$	326 k\$ to 565 k\$	380 k\$ to 646 k\$
Key advantages	Higher operating temperature; reduced cryogenic complexity; fast thermal cycling	Lower conductor cost; mature technology	High J_c margin (likely overkill for 400 mT)
Key challenges	Higher conductor cost; thermal/mechanical design of REBCO stacks	Cryogenic complexity; longer thermal cycling	Brittle conductor; complex fabrication

The cost estimates presented in Table 5 include conductor material, vacuum-compatible coil fabrication with mechanical support structures, thermal management systems, and current leads for each coil. Based on the numerical Biot-Savart field calculations used to scale the current to $B = 400$ mT at the interaction point, the required operating current is approximately $I \approx 1.6$ kA per coil for $N = 100$ turns. This corresponds to an effective current density of $J \approx I/A_{sc} \approx 15.9$ A/mm² for an assumed superconducting area $A_{sc} = 10$ mm $\times$ 10 mm = 100 mm², and an engineering current density of $J_{eng} \approx I/A_{pack} \approx 4.0$ A/mm² referenced to the full conductor pack area $A_{pack} = 20$ mm $\times$ 20 mm = 400 mm². For HTS, an operating current at the 1.6 kA level can be realized using multiple REBCO tapes in parallel, with the required tape count depending on tape grade and operating margin³ [9, 10]. The vacuum chamber and vacuum system costs are excluded, as these are common infrastructure requirements independent of the magnet technology choice. All coils operate within the main vacuum chamber and utilize dry-wound construction with vacuum-compatible insulation materials.

A critical distinction between the HTS and LTS approaches lies in the cryogenic infrastructure requirements. The HTS option operates at liquid-nitrogen temperature (77 K) using closed-cycle cryocoolers with comparatively low technical complexity and rapid thermal cycling. In contrast, the LTS options operate at liquid-helium temperature (4.2 K), requiring more complex cryogenic systems with higher cooling power, longer cooldown times, and increased operational overhead.

Both approaches are compatible with closed-cycle refrigeration suitable for installation in the EIC hadron tunnel environment. The resulting cost differential between 77 K and 4.2 K cryogenic systems is reflected in the infrastructure estimates in Table 5, with typical values of 5 k\$ to 10 k\$ for HTS systems and 50 k\$ to 100 k\$ for LTS systems. Beyond cost, the HTS option offers operational advantages in terms of faster cooldown and recovery, larger temperature margin, and reduced maintenance complexity, which are particularly attractive for a system requiring periodic field shutdowns and reconfiguration.

The comparison suggests that HTS technology at 77 K may offer advantages in terms of operational simplicity and reduced cryogenic infrastructure requirements. In addition, the substantially shorter cool-down times from room temperature to operating conditions (typically 6 h to 12 h for HTS versus 24 h to 72 h for LTS systems) provide a practical operational advantage for commissioning, maintenance, and recovery after interruptions. The compact coil geometry enables Phase 1 deployment in the 150 k\$ to 280 k\$ range depending on technology choice (two coils plus cooling infrastructure), with Phase 2 expansion to full three-axis capability at total system costs in the 330 k\$ to 650 k\$ range.

4.2 Power supply and switching architecture

The baseline operational concept employs one bipolar power supply per coil pair, enabling independent control of each magnetic field axis. For the Phase 1 deployment with B_y generation only, a single power supply rated at approximately 1.6 kA and ± 10 V is sufficient to drive the two-coil system, consistent with the ampere-turn requirement obtained from the numerical Biot-Savart field calculations. Phase 2 expansion to full three-axis capability requires two additional power supplies for the B_x and B_z coil pairs.

An alternative architecture under consideration would use a single high-current power supply shared between the three field axes via a switching matrix that routes current to the selected coil pair. This approach reduces cost and simplifies power distribution, but requires reliable high-current contactors or semiconductor switches capable of operation at currents of order 1.6 kA with

³REBCO: REBa₂Cu₃O_{7-x} (Rare Earth Barium Copper Oxide), where RE = Y, Gd, or other rare earth elements. Also known as 2G (second generation) HTS tape.

minimal resistive losses. The switching logic must ensure that only one coil pair is energized at any given time and must provide galvanic isolation of the inactive coil pairs to prevent unintended circulating currents.

Persistent-mode operation was also considered as a means to improve field stability during the dwell time. In persistent mode, once the target current is reached, a superconducting switch closes to form a closed superconducting loop, after which the external power supply can be disconnected. While this eliminates power-supply drift and ripple, persistent-mode hardware is substantially more mature for low-temperature superconductors than for HTS systems operated near 77 K. Given the modest field stability requirement ($\Delta B/B \sim 1 \times 10^{-3}$) and the relatively short dwell times of order 100 s, persistent-mode operation is not required for the baseline design.

The ramping and switching parameters derived in Sec. 3 imply modest voltage requirements. For a representative ramp time of 5 s, the induced voltage remains at the level of a few volts, well within the capability of standard superconducting magnet power supplies. Quench detection and energy extraction are simplified by the low stored energy of the system (of order 10 kJ per coil pair), which can be safely dissipated using passive protection schemes without the need for complex active dump circuitry.

4.3 Cooling architecture and cryocooler placement

The cryogenic cooling system will be positioned immediately outside the detector acceptance region, at radial distances of approximately 300 mm to 500 mm from the interaction point. This placement minimizes thermal conduction path lengths to the coils while avoiding interference with detector acceptance cones and service access. The configuration requires thermal conduction links between the cryocooler cold heads and the coils, implemented through copper or aluminum thermal straps with appropriate cross-section to handle the thermal load while operating in the vacuum environment.

For HTS operation at 77 K, a closed-cycle cryocooler with cooling capacity of approximately 5 W to 10 W at 77 K is required for the Phase 1 two-coil system, accounting for static heat load from thermal conduction through mechanical supports and dynamic heat load from current leads during operation. Phase 2 expansion to six coils requires a cryocooler with capacity of 15 W to 25 W at 77 K, or multiple smaller units. For LTS operation at 4.2 K, the required cooling capacity is 2 W to 4 W at 4.2 K for the complete six-coil system, despite the lower absolute heat load, due to the significantly reduced Carnot efficiency at liquid helium temperature. Current leads represent the dominant heat load and require thermal intercepts at intermediate temperatures (50 K to 80 K) to minimize the refrigeration load at the operating temperature.

The thermal link design must accommodate the operational cycle of field switching while maintaining coil temperature stability during dwell periods. Vibration isolation is essential to prevent microphonic noise in the silicon detectors. The system is designed for decades-scale operation with serviceable components (cryocooler compressors, cold heads) that can be replaced without warming the coil system or breaking the main vacuum.

For both HTS and LTS options, pulse-tube cryocoolers are the preferred technology, as they avoid moving displacers at the cold head and therefore generate substantially lower mechanical vibration compared to Gifford-McMahon type systems. This is particularly advantageous in the vicinity of the silicon recoil detectors, where microphonic noise and vibration-induced position jitter must be minimized.

4.4 Vendor consultation and procurement strategy

The final technology selection and detailed coil design will be determined in consultation with qualified superconducting magnet vendors. US-based suppliers with relevant experience in compact, cryogen-free superconducting magnet systems include Cryomagnetics, Inc. (Oak Ridge, TN) [11], among others. The procurement strategy will involve a formal request-for-quote (RFQ) process specifying the geometric constraints (Tables 2 and 1), operational parameters (operating current ≈ 1.6 kA per coil, target holding field 400 mT, vacuum compatibility), and performance requirements (field uniformity, ramping and switching times, and stray-field limits).

Vendor input will be essential for finalizing conductor specifications, winding techniques for vacuum-compatible dry-wound coils, thermal management concepts, and quench protection schemes. The RFQ process will be initiated after completion of three-dimensional field simulations, providing vendors with validated field maps and quantitative stray-field requirements relevant to detector operation and nearby instrumentation.

5 Detector system

5.1 Overview and design goals

5.1.1 Beam polarization determination and statistical performance at RHIC HJET

The statistical uncertainty of the beam polarization measurement scales as

$$\frac{\Delta P}{P} = \frac{1}{P A_y \sqrt{N_L + N_R}} \simeq \frac{1}{P A_y \sqrt{N}}, \quad (23)$$

where A_y is the analyzing power, N_L and N_R denote the number of retained events in the left and right detectors, and N denotes the number of retained events per side. This expression defines how the statistical precision of the beam polarization measurement is estimated.

The beam polarization itself is obtained from the ratio of measured beam and target asymmetries,

$$P = \frac{\varepsilon_{\text{beam}}}{\varepsilon_{\text{target}}} Q, \quad (24)$$

with

$$Q = (1 - f_m) Q_{\text{atom}}, \quad (25)$$

where Q_{atom} is the atomic hydrogen polarization of the jet and f_m is the molecular hydrogen fraction⁴. The uncertainty on Q is neglected. The single-spin asymmetry is defined generically as

$$\varepsilon = \frac{N_L - N_R}{N_L + N_R}, \quad (26)$$

and is related to the polarization and analyzing power by

$$\varepsilon_{\text{beam}} = P A_y, \quad \varepsilon_{\text{target}} = Q A_y. \quad (27)$$

⁴Equation (25) is consistent with the RHIC polarization extraction used in the experimental data analysis, as evidenced by the numerical consistency check summarized in Table 6, and is therefore adopted here. Strictly speaking, if f_m denotes the fraction of molecular hydrogen H_2 , each molecule contributing two protons, the effective target polarization would be $Q = (1 - 2f_m) Q_{\text{atom}}$. For $f_m = 0.03$ and $Q_{\text{atom}} = 0.9700$, this would yield $Q \simeq 0.912$, corresponding to a relative change of about 3%. This difference is significant compared to the EIC polarimetry goal of $\Delta P/P < 1\%$ and demonstrates that the molecular hydrogen fraction must be determined with sufficient precision in order to meet the EIC requirements.

Table 6: Numerical consistency check of the beam polarization extraction for the blue beam using RHIC HJET Run 33004. The experimental input parameters and run conditions are summarized below the table.

Quantity	HJET data analysis	From equations
Atomic jet polarization Q_{atom}	0.9700	input
Molecular fraction f_m	0.0300	input
Target polarization $Q = (1 - f_m)Q_{\text{atom}}$	–	0.9409
Target asymmetry $\varepsilon_{\text{target}}$	0.0349 ± 0.0012	input
Beam asymmetry $\varepsilon_{\text{beam}}$	0.0192 ± 0.0012	input
Beam polarization P	0.5171 ± 0.0387	0.517 ± 0.037
Relative uncertainty $\Delta P/P$	0.075	0.072

Data from RHIC HJET Run 33004 (Jan 21, 2022, blue beam) with $A_N = 0.037 \pm 0.0013$, $N = 5.01 \times 10^6$ events retained in the CNI region (from 21.15×10^6 total hits), accumulated over $T_{\text{meas}} = 370$ min. The mean asymmetries were evaluated in the recoil energy range 1.5 MeV to 6.0 MeV. See <https://wiki.bnl.gov/rhicspin/Polarimetry/H-jet/Run22pp>.

Using Eq. (24) and neglecting the uncertainty on Q , the relative uncertainty of the beam polarization is given by

$$\left(\frac{\Delta P}{P}\right)^2 = \left(\frac{\Delta \varepsilon_{\text{beam}}}{\varepsilon_{\text{beam}}}\right)^2 + \left(\frac{\Delta \varepsilon_{\text{target}}}{\varepsilon_{\text{target}}}\right)^2. \quad (28)$$

The consistency of this procedure is illustrated numerically for the blue beam using RHIC HJET Run 33004, as summarized in Table 6.

Representative performance from RHIC HJET operation⁵ demonstrates that, over a measurement duration of $T_{\text{meas}} = 370$ min, a statistically limited beam polarization measurement with a relative precision of about 7.5% was achieved for the blue beam.

The average rate of events retained in the analysis is

$$R_{\text{retained}}^{\text{RHIC}} = \frac{N}{T_{\text{meas}}} = \frac{5.01 \times 10^6}{370 \text{ min}} \approx 226/\text{s}. \quad (29)$$

The events retained in the CNI analysis region constituted approximately 24% of total detector hits, corresponding to an average total detector rate of $R_{\text{total}}^{\text{RHIC}} = \frac{21.15 \times 10^6}{370 \text{ min}} \approx 950/\text{s}$. This total rate includes elastic scattering outside the CNI analysis region, inelastic backgrounds, and accidental coincidences. The proposed compact detector geometry with enhanced solid angle acceptance (Sec. 2.2) is expected to provide comparable or improved selection efficiency while substantially increasing the rate of events retained in the analysis through increased geometric acceptance.

5.1.2 Events per bunch crossing at RHIC and EIC

The RHIC HJET target provides an atomic hydrogen flux of $\Phi = 12.4(2) \times 10^{16}$ atoms/s with a jet target thickness along the beam of $1.3(2) \times 10^{12}$ atoms/cm² and a beam diameter of approximately 10 mm [1]. With the RHIC beam parameters listed in Table II of Ref. [1], the number of protons per bunch is $N_p = 2.0 \times 10^{11}$, circulating at a revolution frequency of $f_{\text{rev}} = 78.175$ kHz with $N_b = 120$ bunches, corresponding to a bunch crossing frequency of $f_b = N_b f_{\text{rev}} = 9.381$ MHz.

⁵RHIC HJET Run 33004 (Jan 21, 2022, blue beam) with $A_N = 0.037 \pm 0.0013$ using $N = 5.01 \times 10^6$ events retained in the CNI analysis region (from 21.15×10^6 total detector hits) accumulated over measurement duration $T_{\text{meas}} = 370$ min. Data from <https://wiki.bnl.gov/rhicspin/Polarimetry/H-jet/Run22pp>.

The average number of detector hits per bunch crossing can be estimated from the total detector rate and the bunch crossing frequency:

$$n_{\text{evt/crossing}}^{\text{RHIC}} = \frac{R_{\text{total}}^{\text{RHIC}}}{f_b^{\text{RHIC}}} = \frac{950/\text{s}}{9.381 \text{ MHz}} \approx 1.0 \times 10^{-4}. \quad (30)$$

This corresponds to approximately one detector hit per 10^4 bunch crossings. The average time between detector hits at RHIC is

$$\Delta t_{\text{avg}}^{\text{RHIC}} = \frac{1}{R_{\text{total}}^{\text{RHIC}}} = \frac{1}{950/\text{s}} \approx 1 \text{ ms}. \quad (31)$$

The low occupancy per crossing indicates that pile-up from multiple interactions within a single bunch crossing is negligible under typical RHIC HJET operating conditions, and bunch assignment is unambiguous given the large separation between detector hits relative to the bunch spacing of $\tau_b = 106.6 \text{ ns}$.

For EIC, following the logic from Eq. (30), we obtain

$$n_{\text{evt/crossing}}^{\text{EIC}} = \frac{R_{\text{total}}^{\text{EIC}}}{f_b^{\text{EIC}}} = \frac{3500/\text{s}}{90.683 \text{ MHz}} \approx 3.9 \times 10^{-5}. \quad (32)$$

The average time between detector hits is

$$\Delta t_{\text{avg}}^{\text{EIC}} = \frac{1}{R_{\text{total}}^{\text{EIC}}} = \frac{1}{3500/\text{s}} \approx 0.29 \text{ ms}. \quad (33)$$

5.1.3 Recoil kinematics and detector footprint in the CNI region

For the proposed EIC HJET system, several operational parameters differ significantly from RHIC conditions. The EIC beam parameters listed in Table II of Ref. [1] show a higher average beam current ($I_{\text{avg}} = 1.002 \text{ A}$ at injection, 1.003 A at flattop) compared to RHIC (0.301 A), with bunch spacing reduced to $\tau_b = 11.027 \text{ ns}$ (flattop) compared to 106.6 ns at RHIC. Assuming comparable target parameters and elastic scattering cross sections, the total detector rate is expected to scale approximately with the beam current ratio, suggesting rates in the range of $3000/\text{s}$ to $3500/\text{s}$ for the EIC configuration.

The compact detector geometry with silicon modules positioned at $r = 220 \text{ mm}$ (compared to 750 mm at RHIC) substantially increases the geometric solid angle acceptance. However, as discussed in Sec. 2.2, the CNI region corresponds to a relatively narrow angular range, and the elastic scattering acceptance does not scale with the full solid angle factor $(750/220)^2$.

To quantify the angular extent of the CNI region on the detector surface, we calculate the recoil-proton lab-frame angle from complete 4-momentum conservation in fixed-target elastic pp scattering. For the momentum transfer values $|t|$ relevant to the CNI region, the recoil proton emerges at angles close to, but measurably less than, perpendicular to the beam direction.

For a beam proton with 4-momentum $p_1 = (E_{\text{beam}}, p_{\text{beam}}, 0, 0)$ incident on a target proton at rest with $p_2 = (m_p, 0, 0, 0)$, the recoil proton 4-momentum $p_4 = (E_4, \vec{p}_4)$ is determined by energy-momentum conservation $p_1 + p_2 = p_3 + p_4$, where p_3 denotes the scattered beam proton. The recoil kinetic energy is $T_R = |t|/(2m_p)$ with $E_4 = m_p + T_R$ and recoil momentum magnitude

$$p_R = \sqrt{E_4^2 - m_p^2}.$$

The recoil angle θ_R with respect to the beam axis follows from momentum conservation in the transverse and longitudinal directions. For the scattered beam proton at angle θ_3 , determined from the Mandelstam invariant $t = (p_1 - p_3)^2$, transverse momentum conservation yields

$$\sin \theta_R = \frac{p_3 \sin \theta_3}{p_R}, \quad (34)$$

with $p_3 = \sqrt{E_3^2 - m_p^2}$ and $E_3 = E_{\text{beam}} + m_p - E_4$. This relation defines the recoil angle without approximation.

In the CNI momentum transfer region $|t| \sim 1 \times 10^{-3}$ to $2 \times 10^{-2} (\text{GeV}/c)^2$, the recoil proton emerges at angles in the range $\theta_R \approx 85^\circ$ to 89° with respect to the beam direction. The recoil angle θ_R depends on both the momentum transfer $|t|$ and the beam energy, and is listed in Table 7 for representative $|t|$ values at beam energies of 23.5 GeV and 275 GeV.

At the detector distance of $r = 220$ mm (perpendicular to the beam axis), recoil protons at angle θ_R from the beam axis intersect the detector plane at longitudinal position

$$z = \frac{r}{\tan \theta_R}. \quad (35)$$

Since the recoil angles are less than 90° , the recoil protons carry a small forward momentum component and hit the detector downstream from the interaction point at positive z . The CNI region spans longitudinally from $z \approx 4$ mm to $z \approx 21$ mm on the detector surface, with the center located at $z \approx 12$ mm downstream from the interaction point, as shown in Table 7. The longitudinal extent $\Delta z \approx 17$ mm is approximately the same at both beam energies because the recoil angle is primarily determined by the transverse momentum transfer $p_\perp \approx \sqrt{|t|}$, which is independent of beam energy.

While this extent is modest compared to the $100 \text{ mm} \times 100 \text{ mm}$ active detector area, it is not negligible and should be considered in detector positioning and acceptance calculations. The primary acceptance gain of the compact geometry thus arises from increased azimuthal coverage rather than polar angular acceptance, resulting in an effective enhancement factor of approximately 6-7 for elastic events in the CNI region (Table 3). In contrast, inelastic background processes with broader angular distributions may scale more closely with the full geometric factor $(750/220)^2 \approx 12$, leading to a less favorable signal-to-background ratio compared to RHIC operation.

Most critically, the reduced bunch spacing of 11 ns becomes comparable to the recoil-proton time-of-flight range of 4 ns to 22 ns (Table 7), introducing potential ambiguities in bunch assignment even at low occupancy per crossing. A bunch-passage time reference t_0 can be provided by the machine timing/RF system (phase-locked bunch clock) distributed to the detector readout; no dedicated local start detector at the target is required, provided the latency is calibrated and stable.

5.1.4 Timing requirements and bunch association

The primary timing requirement of the recoil detector system is unambiguous association of each detected recoil proton with the correct bunch crossing of the hadron beam in the polarized target. This requirement is driven by the short bunch spacing, $\tau_b \simeq 11$ ns, together with the finite temporal bunch width, $\sigma_t \simeq 0.2$ ns (rms), and recoil-proton time-of-flight values that can exceed one bunch interval for low momentum transfer $|t|$ in the CNI region (see Table 7). The relevant bunch parameters for EIC injection and flattop operation are summarized in Table II of Ref. [1].

⁶CSDA: Continuous Slowing Down Approximation

Table 7: Representative recoil-proton parameters for elastic pp scattering in the CNI region. The recoil kinetic energy is computed from $T_R = |t|/(2m_p)$ with $m_p = 0.938\,272\,\text{GeV}/c^2$, and the recoil momentum and velocity are obtained from standard relativistic kinematics. The recoil scattering angle θ_R from the beam axis is calculated from complete 4-momentum conservation for fixed-target collisions at beam energies of 23.5 GeV and 275 GeV via Eq. (34). The longitudinal position z represents the intersection point of the recoil trajectory with the detector plane at perpendicular distance $r = 220\,\text{mm}$, calculated via Eq. (35). The time of flight is evaluated for a straight-line path of length $L = 220\,\text{mm}$. The CSDA⁶ range in silicon is interpolated from the NIST PSTAR database [12].

$ t $ [(GeV/c) ²]	T_R [MeV]	p_R [MeV/c]	β	θ_R (23.5 GeV) [°]	z (23.5 GeV) [mm]	θ_R (275 GeV) [°]	z (275 GeV) [mm]	t_{TOF} [ns]	CSDA range in Si [μm]
0.001	0.532 89	31.627	0.033 689	88.995	3.86	89.031	3.72	21.783	6.7
0.002	1.065 80	44.734	0.047 623	88.579	5.46	88.630	5.27	15.409	18.1
0.003	1.598 70	54.796	0.058 301	88.260	6.68	88.322	6.44	12.587	33.7
0.005	2.664 50	70.761	0.075 203	87.754	8.63	87.835	8.32	9.758	76.4
0.010	5.328 90	100.140	0.106 130	86.825	12.20	86.939	11.76	6.915	242.6
0.015	7.993 40	122.740	0.129 700	86.113	14.95	86.253	14.41	5.658	485.4
0.020	10.658 0	141.820	0.149 460	85.514	17.26	85.675	16.64	4.910	799.6
0.030	15.987 0	173.940	0.182 280	84.511	21.14	84.708	20.38	4.026	1623.0

A bunch-synchronous reference time t_0 is provided by the machine timing system, phase-locked to the RF and distributed to the detector readout with calibrated and stable latency. This reference time is equivalent to a local bunch-passage detector at the target and defines the nominal interaction time for each bunch crossing.

For each recoil-proton hit, the detector measures a hit time t_{hit} and forms the time difference

$$\Delta t = t_{\text{hit}} - t_0. \quad (36)$$

Because the recoil-proton time of flight t_{TOF} can exceed τ_b , the measured Δt alone does not uniquely determine the originating bunch index. The correct bunch assignment is obtained by exploiting the strong correlation between recoil kinetic energy T_R and time of flight. For each candidate bunch index n , the condition

$$\Delta t + n \tau_b \simeq t_{\text{TOF}}(T_R, L_{\text{path}}) \quad (37)$$

must be satisfied within the combined timing and energy resolution, where

$$t_{\text{TOF}}(T_R, L_{\text{path}}) = \frac{L_{\text{path}}}{\beta c}, \quad (38)$$

with

$$\beta = \frac{p_R}{\sqrt{p_R^2 + m_p^2}}, \quad (39)$$

and p_R inferred from the measured recoil energy T_R .

Given the very low event occupancy per bunch crossing, pile-up from multiple interactions in a single crossing is negligible. Consequently, the timing requirement is not driven by hit separation or pile-up rejection, but solely by the need to resolve the integer bunch-offset ambiguity. Timing resolution at the level of a few nanoseconds, referenced to the machine bunch clock, is sufficient

to achieve high bunch-assignment fidelity when combined with recoil-energy information. Sub-nanosecond timing improves robustness and diagnostic capability but is not a strict requirement for correct bunch identification.

An optional dedicated fast timing layer (e.g. diamond or LGAD-based) may be employed to improve timing precision and provide redundancy. However, the baseline detector concept does not rely on sub-nanosecond timing performance.

Monte Carlo inputs and performance metrics. The detector timing performance is evaluated in simulation using the following ordered set of inputs:

1. Bunch structure with bunch spacing $\tau_b = 11$ ns and known bunch reference times t_0 , provided by the machine timing system phase-locked to the RF and distributed to the detector readout, where the intrinsic bunch arrival time is smeared by the rms temporal bunch width σ_t as listed in Table II of Ref. [1].

2. **Interaction vertex distribution:** The recoil interaction vertex (x_0, y_0, z_0) is sampled within the finite HJET target volume. The longitudinal coordinate is drawn from a uniform distribution

$$z_0 \in [-5, +5] \text{ mm}, \quad (40)$$

corresponding to a total target length $\Delta z \simeq 10$ mm. The transverse coordinates are sampled from Gaussian distributions,

$$x_0 \sim N(0, \sigma_x^2), \quad y_0 \sim N(0, \sigma_y^2), \quad (41)$$

using the EIC flattop beam sizes (σ_x, σ_y) listed in Table II of Ref. [1]. A uniform azimuthal recoil direction $\phi \in [0, 2\pi)$ is assumed, and for timing studies the recoil polar angle is taken as $\theta_R = 90^\circ$. This represents a simplification appropriate for the TOF geometry calculation, as the actual CNI recoil angles lie in the range 85° – 89° as shown in Table 7. The sampled vertex and azimuth define the geometric flight path to the cylindrical detector surface and are used to compute the event-by-event time of flight

$$t_{\text{TOF}} = \frac{L}{\beta c}. \quad (42)$$

The associated interaction-time spread due to the finite target length,

$$\Delta t_z \simeq \frac{\Delta z}{\beta_{\text{beam}} c} \approx 33 \text{ ps}, \quad (41)$$

is negligible compared to ns-scale detector timing.

3. Recoil-proton kinematics sampled in the CNI region, including the recoil kinetic energy T_R , momentum p_R , and velocity β , derived from relativistic kinematics. The corresponding time of flight t_{TOF} is computed from the event-by-event geometric flight path and the recoil velocity.
4. Detector hit-time resolution parameterized by a Gaussian smearing with width σ_{det} , representing the combined effects of sensor signal formation and front-end electronics. Multiple values are explored in simulation ($\sigma_{\text{det}} = 1$ ns, 2 ns, and 5 ns) to assess the robustness of bunch-assignment performance against realistic variations in detector technology, sensor thickness, segmentation, and readout bandwidth.

5. Energy measurement resolution for the recoil proton, from which the recoil momentum p_R and velocity β are inferred. In the present timing-only simulation, the measured recoil energy is used to predict the expected time of flight for bunch-assignment consistency checks; detailed magnetic-field transport and hit-position reconstruction will be treated in dedicated tracking studies.

The primary performance metric is the probability of correct and unambiguous bunch assignment as a function of $|t|$ across the CNI region. The design goal is to achieve correct bunch association for more than 95% of selected elastic recoil events over the full $|t|$ range relevant for polarization analysis.

5.2 Physics observables and detector requirements

The recoil detector system must measure elastic pp scattering in the CNI region with sufficient precision to extract the beam polarization from azimuthal asymmetries. The relevant physics observables are the recoil kinetic energy T_R , the azimuthal recoil angle ϕ , and the hit time of the recoil proton referenced to the machine bunch clock.

The recoil kinetic energy T_R determines the momentum transfer via

$$|t| \simeq 2m_p T_R, \quad (42)$$

and thus defines the $|t|$ bin used for the analyzing power $A_y(|t|)$ [2]. Accurate and stable measurement of T_R is therefore required over the recoil energy range corresponding to the CNI region. The determination of $|t|$ relies solely on the recoil energy and is independent of the recoil angle.

The role of the azimuthal angle ϕ in the polarization measurement is clarified by considering the expected event yield in the recoil detector. For vertically polarized protons, with the polarization vector aligned along $\vec{e}_y$ and the beam direction along $\vec{e}_z$, the number of detected elastic recoil events in a given momentum-transfer bin $|t|$ and azimuthal angle ϕ can be written as

$$N(|t|, \phi) = N_0(|t|) a(|t|, \phi) \eta(|t|, \phi) [1 + P A_y(|t|) \cos \phi], \quad (43)$$

where $N_0(|t|)$ is the ϕ -integrated yield in the $|t|$ bin, $a(|t|, \phi)$ denotes the geometrical acceptance, $\eta(|t|, \phi)$ represents the detection and reconstruction efficiency, P is the beam polarization, and $A_y(|t|)$ is the analyzing power for fixed-target elastic pp scattering in the CNI region. The azimuthal angle ϕ is defined in the transverse plane, measured from the $\vec{e}_x$ axis toward the $\vec{e}_y$ axis. In this coordinate convention, vertical polarization leads to a left and right asymmetry along $\vec{e}_x$, which gives rise to the $\cos \phi$ modulation in Eq. (43). The experimentally measured left-right asymmetries are defined via Eqs. (26) and (27), where L and R denote the yields integrated over the left and right hemispheres after acceptance and efficiency corrections.

In practice, the azimuthal angle ϕ is determined from the impact position of the recoil proton on the detector surface. Specifically, ϕ is defined by the first detector element encountered by the recoil proton, which fixes the azimuthal hit position and acceptance. The detector must therefore provide sufficient spatial resolution and geometrical stability at this first measurement stage to control azimuthal acceptance and related systematic effects. Reconstruction of the polar recoil angle is not a primary requirement, since the CNI kinematics are fully determined by the recoil energy T_R , while the azimuthal angle ϕ enters only through the polarization-dependent asymmetry.

In addition, the detector system must provide timing information referenced to the machine bunch clock, as discussed in Sec. 5.3, in order to associate each recoil event unambiguously with the correct hadron bunch crossing. This requirement is driven by the short bunch spacing and the recoil-proton time-of-flight spread, rather than by detector occupancy or pile-up considerations. The

timing information must be sufficient to resolve the integer bunch-offset ambiguity when combined with recoil energy information.

The detector system must operate in the presence of magnetic holding fields up to 400 mT in all required configurations, including $\pm B_x$, $\pm B_y$, and $\pm B_z$. While the magnetic field does not affect the recoil kinetic energy T_R , it modifies the recoil-proton trajectory and the corresponding hit position on the detector surface. The detector geometry, segmentation, and reconstruction logic must therefore support consistent azimuthal angle reconstruction and acceptance definition for all holding-field orientations.

These requirements define the functional performance of the recoil detector system and provide the basis for the detector concept, geometry, and readout architecture discussed in the following sections.

5.3 Timing requirements and bunch association

This subsection summarizes the resulting detector timing requirements, independent of the specific Monte Carlo implementation discussed in Sec. 5.1.4.

The recoil detector system must provide timing information referenced to the machine bunch clock in order to associate each detected recoil proton unambiguously with the correct hadron bunch crossing. This requirement is driven by the short bunch spacing in the hadron storage ring and by the wide recoil-proton time-of-flight distribution in the CNI region, rather than by detector occupancy or pile-up considerations.

A bunch-synchronous reference time t_0 is provided by the machine timing system, phase-locked to the RF and distributed to the detector readout with calibrated and stable latency. For each recoil event, the detector measures a hit time t_{hit} and forms the time difference given in Eq. (36).

For recoil protons in the CNI region, the recoil momentum satisfies $p_R \simeq \sqrt{|t|}$, and the corresponding velocity is given by β via Eq. (39). For a recoil trajectory of length L_{path} , determined by the detector geometry, the expected time of flight is given by Eq. (38). Over the CNI momentum-transfer range, the recoil-proton time of flight t_{TOF} spans several nanoseconds and can exceed one bunch spacing interval, as quantified by the values listed in Table 7.

Because $t_{\text{TOF}} \gtrsim \tau_b$ for low $|t|$, the measured time difference Δt alone does not uniquely identify the originating bunch crossing. For a given recoil event, multiple candidate bunch indices n must therefore be considered, corresponding to the condition defined in Eq. (37). Correct bunch assignment is achieved by selecting the unique integer n for which this condition is satisfied within the combined timing and energy resolution.

Given the very low event occupancy per bunch crossing, pile-up from multiple interactions in a single crossing is negligible. Consequently, the timing requirement is not driven by hit separation, but solely by the need to resolve the integer bunch-offset ambiguity. Timing performance at the level of a few nanoseconds, referenced to the machine bunch clock, is sufficient to achieve high bunch-assignment fidelity when combined with recoil energy information.

5.4 Operation in magnetic holding fields

The recoil detector system must operate in the presence of magnetic holding fields required for polarized-target operation. Field configurations include all six orientations $\pm B_x$, $\pm B_y$, and $\pm B_z$, with field magnitudes up to 400 mT in the interaction region.

The magnetic holding field does not affect the recoil kinetic energy T_R and therefore does not modify the determination of the momentum transfer $|t|$, because the Lorentz force acts perpendicular to the particle velocity and performs no work; this reflects energy conservation associated with

time-translation invariance in a static magnetic field. However, it alters the recoil-proton trajectory between the interaction point and the detector surface, leading to field-dependent shifts of the hit position and azimuthal mapping.

The detector geometry, segmentation, and reconstruction logic must therefore support consistent azimuthal angle reconstruction and acceptance definition for all holding-field orientations. In particular, the mapping between detector hit position and azimuthal angle ϕ must be well defined and reproducible for each field configuration.

Magnetic field effects must be incorporated in dedicated detector simulations and reconstruction studies through validated three-dimensional field maps. These maps provide the basis for recoil-proton transport, acceptance modeling, and consistency checks between measured hit positions and expected recoil trajectories. The required field maps are derived from numerical Biot-Savart calculations and have been validated for field magnitude and uniformity in the interaction region, as described in Sec. 3.7.

Operation in magnetic fields further requires that detector sensors, front-end electronics, and readout components maintain stable performance in the presence of the holding field, without degradation of timing, energy measurement, or spatial response. Table 8 summarizes the magnetic field magnitude at the recoil-detector locations for the three holding-field configurations, evaluated at representative detector-center positions at a radius of $r \approx 220$ mm. While the holding field reaches 400 mT in the interaction region, the field experienced by the silicon detectors is significantly reduced and depends on the field orientation and detector location. For the transverse-field configurations (B_x and B_y), the detector planes aligned with the corresponding field axis experience fields at the level of ~ 50 mT, while the orthogonal detector planes see fields below 10 mT. For the longitudinal configuration (B_z), the field magnitude at all detector positions remains below 6 mT.

Table 8: Magnetic field magnitude at the detector radius ($r \approx 220$ mm) for the three holding-field configurations, obtained from the numerical field maps. For each configuration, the quoted values correspond to representative detector-center positions on the $\pm\vec{e}_x$ and $\pm\vec{e}_y$ axes at $(x, y, z) = (\pm 220, 0, 0)$ and $(0, \pm 220, 0)$, respectively.

Configuration	$ \vec{B} $ at $(\pm 220, 0, 0)$ [T]	$ \vec{B} $ at $(0, \pm 220, 0)$ [T]
B_x pair	0.050506	0.007029
B_y pair	0.007029	0.050506
B_z pair	0.005852	0.005852

The magnetic field strengths at the detector locations are well within the operational envelope of modern segmented silicon sensors and front-end electronics. Field reversal between $\pm B_x$, $\pm B_y$, and $\pm B_z$ configurations does not introduce fundamental limitations for detector operation, as the field magnitude at the sensors remains modest and the associated trajectory distortions are fully accounted for through field-map-based reconstruction. Consequently, operation in all required holding-field configurations is compatible with stable detector performance and controlled systematic uncertainties.

5.5 Recoil detector concept

The recoil detector system is conceived as a compact, segmented silicon detector optimized for elastic pp scattering in the CNI region. The concept follows directly from the physics observables and detector requirements summarized in Sec. 5.2 and from the timing and bunch-association constraints discussed in Sec. 5.3.

The baseline layout employs two radially stacked detector layers arranged around the beam axis at a distance of approximately $r = 220$ mm from the interaction region. The first detector layer provides the azimuthal hit assignment relevant for the polarization analysis through the location of the first intercepted detector element, while the recoil kinetic energy T_R is reconstructed from the combined energy deposition in both layers, within a compact geometry compatible with operation in magnetic holding fields.

5.5.1 Energy measurement and azimuthal acceptance

The azimuthal segmentation of the detector defines the acceptance entering the polarization asymmetry measurement discussed in Sec. 5.2. Stable geometry and uniform response of the detector elements are therefore essential for controlling azimuthal acceptance effects and related systematic uncertainties.

The recoil kinetic energy T_R is reconstructed from the total energy deposited in the detector system using the combined response of both layers. This increases the effective sensitive thickness and improves energy resolution without requiring polar-angle reconstruction, which is not needed for CNI kinematics. The two-layer configuration also provides redundancy for hit validation and background suppression.

5.5.2 Sensor technology and readout considerations

The detector concept is compatible with commercially available segmented silicon sensors with fast signal formation and good radiation tolerance. As a reference implementation, silicon pad or strip sensors as specified in the Hamamatsu Photonics sensor data sheets (e.g., S13804 series [13]) provide suitable segmentation, thickness options, and timing performance for this application. The final sensor choice will be guided by energy resolution, timing performance, and integration with the readout electronics.

The detector segmentation and readout architecture are designed to support reliable bunch association using hit-time information referenced to the machine bunch clock, as described in Sec. 5.3. Sub-nanosecond intrinsic sensor timing is not a strict requirement for the baseline concept, but faster sensors may be employed to improve robustness and diagnostic capability.

The baseline detector module design is based on single-sided silicon strip detectors of the Hamamatsu S13804 series [14]. Each sensor provides a large active area and a two-column strip layout, which naturally supports the two-layer module concept used here. The first layer is oriented to measure the azimuthal hit coordinate, while the second layer is oriented perpendicular to the first to provide an independent spatial coordinate for hit validation and background suppression (see Fig. 1). Representative sensor parameters relevant for geometry definition and readout channel count are summarized in Table 9.

With 1024 strips per sensor layer, the two-layer detector module comprises 2048 readout channels per module, without any ganging or multiplexing.

5.6 Detector readout and timing

The recoil detector readout system is conceived as a streaming system that continuously records time-stamped detector hits together with the corresponding energy deposition and hit position. This streaming approach follows the standard data-acquisition philosophy adopted for the EIC main detectors and the polarimetry systems. The readout is synchronized to the accelerator timing system and operates with a stable reference to the machine bunch clock, as required by the analysis framework established in Sec. 5.3.

Table 9: Key parameters of the Hamamatsu S13804 single sided silicon strip detector used as the baseline sensor choice for the recoil detector modules (from the Hamamatsu S13804 datasheet, December 2023) [14].

Parameter	Value
Sensor type	Single sided SSD, PolySi bias AC readout
Si thickness	$320\text{ }\mu\text{m} \pm 15\text{ }\mu\text{m}$
Chip size	$(98.770\text{ mm} \pm 0.020\text{ mm}) \times (98.770\text{ mm} \pm 0.020\text{ mm})$
Active area	$97.280\text{ mm} \times 97.280\text{ mm}$
Strip layout	512 channels $\times$ 2 columns
Number of strips	1024
Strip pitch	$190\text{ }\mu\text{m}$
Strip implant width	$80\text{ }\mu\text{m}$
Strip readout Al width	$90\text{ }\mu\text{m}$
Readout pad size	$165\text{ }\mu\text{m} \times 100\text{ }\mu\text{m}$ (per pad)
Maximum reverse voltage	200 V
Full depletion voltage	$\leq 100\text{ V}$
Defective strip rate	$\leq 5\%$
Operating temperature	-20°C to 60°C

For each detector channel, the readout provides digitized energy and timing information referenced to the distributed bunch-synchronous clock signal. The data stream is processed without a hardware trigger, reflecting the very low event occupancy and allowing flexible event selection, monitoring, and reconstruction at the software level.

The readout architecture must ensure stable timing alignment, low jitter with respect to the machine reference, and reproducible timing behavior over extended data-taking periods. Timing information is used together with the reconstructed recoil kinetic energy and detector geometry in the analysis chain to support event reconstruction for bunch association and quality control.

The detector readout must operate reliably at the expected hit rates and in the presence of magnetic holding fields in all required configurations $\pm B_x$, $\pm B_y$, and $\pm B_z$. While the central holding field reaches up to 400 mT in the interaction region, the magnetic field magnitude at the detector radius ($r \approx 220\text{ mm}$) is significantly lower and depends on the active coil pair, as summarized in Table 8. The front-end electronics and readout chain must therefore maintain stable timing, energy measurement, and efficiency in magnetic fields at the detector location at the level of a few to $\sim 50\text{ mT}$, independent of field orientation and polarity.

Timing performance at the level of a few nanoseconds is sufficient for the intended analysis when combined with recoil energy information. Faster intrinsic timing resolution may be implemented to improve robustness, monitoring capability, and diagnostic flexibility, but is not a strict requirement of the baseline detector concept.

5.7 Simulation and performance validation

The performance of the recoil detector system is evaluated using Monte Carlo simulations at different levels of detail, including dedicated studies of timing and bunch association as well as ongoing full-geometry simulations incorporating validated three-dimensional magnetic field maps described in Sec. 3.7. The simulation provides a consistent framework for studying recoil-proton transport, detector acceptance, and the mapping between interaction kinematics and measured observables.

Elastic pp scattering events are generated in the CNI region according to the expected momentum-transfer distribution. Depending on the study, recoil protons are either transported through the magnetic holding field to the detector surface or treated with simplified geometry for timing-performance evaluations. The recoil kinetic energy, hit position, and time-of-flight are recorded at the detector layers, allowing direct comparison with the reconstruction strategy outlined in Sec. 5.5 and Sec. 5.6.

The magnetic field maps are used without approximation to transport recoil protons and to evaluate field-dependent acceptance effects for all holding-field configurations $\pm B_x$, $\pm B_y$, and $\pm B_z$. This ensures that azimuthal angle reconstruction, detector acceptance, and efficiency corrections are treated consistently across field orientations.

The primary performance metrics include the reconstructed $|t|$ resolution, azimuthal acceptance uniformity, sensitivity to left-right asymmetries, and robustness against field-dependent trajectory distortions. These studies confirm that the detector concept satisfies the physics requirements for polarization extraction in the CNI region and provides adequate control of systematic effects associated with geometry, timing, and magnetic fields.

The simulation framework also serves as the basis for optimization studies of detector segmentation, layer thickness, and operating parameters, and will be used to validate reconstruction algorithms and systematic-uncertainty estimates prior to deployment.

5.7.1 Toy Monte Carlo of bunch assignment performance

A toy Monte Carlo simulation was developed to quantify the performance of recoil-bunch assignment based on the internal consistency between recoil time-of-flight and recoil energy. The purpose of this study is to determine under which timing conditions the correct beam bunch can be identified unambiguously, and how this capability depends on recoil kinematics and detector timing resolution under representative EIC operating conditions.

The simulation generates a total of 10^6 independent recoil-proton events, as summarized in Table 10. Each event is assigned a true momentum transfer $|t|$, drawn uniformly within predefined $|t|$ intervals (“ $|t|$ bins”) (see Table 7). These bins are fixed a priori and are analyzed independently. This binning is essential because the recoil time-of-flight depends strongly on $|t|$, so both the timing structure and the ambiguity of the bunch assignment vary significantly across the kinematic range. Evaluating the results separately in each $|t|$ bin makes this dependence explicit and avoids mixing regions with fundamentally different timing behavior.

For each generated event, the ideal recoil time-of-flight t_{TOF} is computed from the recoil kinematics and a fixed effective flight path, as defined earlier. A measured detector hit time stamp is then formed by adding statistically independent timing contributions associated with the beam bunch structure and the detector response,

$$t_{\text{hit}} = i \tau_b + t_{\text{TOF}} + \delta t_b + \delta t_{\text{det}} , \quad (44)$$

where i denotes the unknown true bunch index, τ_b is the bunch spacing, δt_b represents the intrinsic temporal spread of the beam bunch, δt_{det} is the detector timing uncertainty characterized by a resolution σ_{det} .

The bunch-assignment algorithm tests a discrete set of candidate integer offsets

$$n = 0, \dots, n_{\text{max}} , \quad (45)$$

corresponding to the number of bunch periods by which the measured hit time may be shifted relative to the true production bunch. For each candidate offset, a residual is formed by comparing

the reconstructed hit time to the predicted recoil time-of-flight,

$$r_n = (t_{\text{hit}} \bmod \tau_b) + n \tau_b - t_{\text{TOF,pred}}, \quad (46)$$

where $t_{\text{TOF,pred}}$ is the recoil time-of-flight predicted from the reconstructed recoil energy and the assumed flight path length, using the same kinematic relations as for the ideal time-of-flight.

The candidate offset that minimizes the absolute residual $|r_n|$ is selected as the best bunch hypothesis for the event. An event is accepted if this minimum residual satisfies a consistency requirement expressed in units of the effective timing resolution σ_{eff} ,

$$|r_n| < k \sigma_{\text{eff}}, \quad (47)$$

where k is a dimensionless acceptance parameter. The effective timing resolution is defined as the quadrature sum of the relevant timing contributions,

$$\sigma_{\text{eff}} = \sqrt{\sigma_{\text{det}}^2 + \sigma_b^2}, \quad (48)$$

with σ_{det} denoting the detector timing resolution and σ_b the rms bunch time spread. In the present study, $k = 5$ is used to ensure high acceptance while maintaining excellent purity in bunch assignment. The parameter k controls the width of the acceptance window in units of σ_{eff} . Larger values of k increase the acceptance probability but allow a higher rate of incorrect bunch assignments, while smaller values of k improve purity at the cost of reduced efficiency. The detector timing resolution σ_{det} enters directly into σ_{eff} and therefore represents a key performance parameter explored in this study.

Table 10: Main parameters used in the toy Monte Carlo study of recoil–bunch assignment. EIC timing parameters correspond to representative flattop conditions as discussed in Ref. [1].

Parameter	Symbol / value	Comment
Number of simulated events	$N = 10^6$	Toy MC statistics
Bunch spacing (EIC)	$\tau_b = 11.027 \text{ ns}$	EIC flattop
Bunch temporal width (EIC)	$\sigma_b = 0.2 \text{ ns}$	rms bunch time spread
Detector timing resolution	$\sigma_{\text{det}} = \{0.5, 1, 2, 3, 4, 5\} \text{ ns}$	discrete scan values
Consistency window factor	$k = 5$	acceptance criterion $ r_n < k \sigma_{\text{eff}}$
Maximum tested bunch offset	$n_{\text{max}} = 5$	candidate bunch search range
Effective recoil flight path	$L = 220 \text{ mm}$	toy geometry parameter
$ t $ binning range	$0.0005\text{--}0.035 \text{ (GeV}/c)^2$	non-uniform bins, CNI region
Longitudinal vertex range	$z_0 \in [-5, +5] \text{ mm}$	uniform HJET target length (timing study)
Transverse vertex distribution	$x_0 \sim \mathcal{N}(0, \sigma_x^2), y_0 \sim \mathcal{N}(0, \sigma_y^2)$	EIC flattop beam sizes (timing-only geometry)
Azimuthal recoil angle	$\phi \in [0, 2\pi)$	uniform illumination
Polar recoil angle	$\theta_R = 90^\circ$	fixed; mrad-scale deviations neglected

The inclusion of longitudinal and transverse vertex smearing accounts for event-by-event variations in recoil path length relevant for timing; their net effect on the bunch-assignment performance is small compared to the detector timing resolution and is therefore not resolved separately in the figures shown.

The behavior of the bunch-assignment algorithm is illustrated using a sequence of diagnostic distributions derived from the toy Monte Carlo simulation. The figures are ordered to reflect the logical progression from the underlying kinematic structure of the problem to the resulting assignment performance. We first establish the relation between recoil kinematics and time-of-flight relative to the bunch structure, then examine how the timing-consistency criterion acts on individual events, and finally present the resulting bunch-assignment efficiency as the primary performance observable.

Figure 5 summarizes the kinematic input to the bunch-assignment problem by showing the recoil-proton time of flight as a function of the momentum transfer $|t|$, together with reference lines at integer multiples of the bunch spacing τ_b . For a given recoil kinematics, this representation directly shows how many bunch periods the recoil time of flight can span. The largest number of intersections between the $t_{\text{TOF}}(|t|)$ curve and the lines at $m\tau_b$ determines the maximum number of bunch offsets that can contribute within the kinematic range of interest. This observation provides a direct and transparent criterion for choosing the maximum offset n_{max} used in the bunch-assignment algorithm and establishes the connection between recoil kinematics and the discrete timing structure of the accelerator.

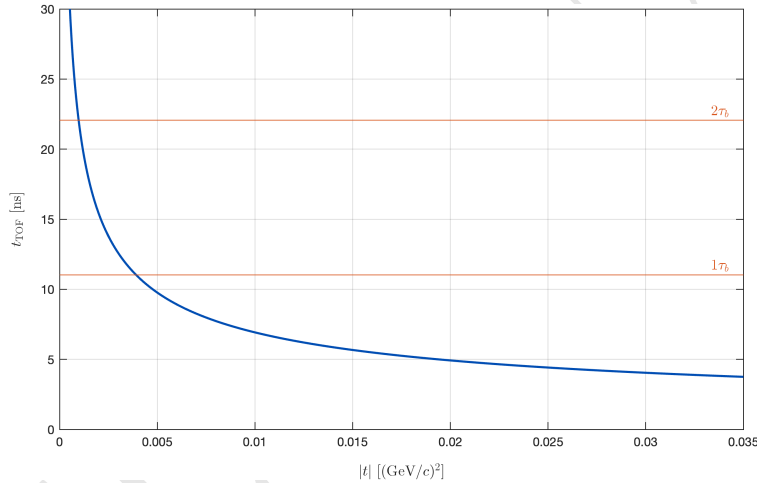


Figure 5: Recoil-proton time of flight as a function of momentum transfer $|t|$. Horizontal lines indicate integer multiples of the bunch spacing τ_b .

The effect of the time-of-flight consistency requirement is illustrated in Fig. 6, which shows the distributions of the residual r_n for accepted events in representative low- and intermediate- $|t|$ bins. The residuals are narrowly centered around zero, demonstrating that the selected bunch hypothesis is consistent with the recoil kinematics within the allowed timing uncertainty. The observed width of the distributions is governed by the effective timing resolution σ_{eff} , while incorrect bunch hypotheses populate broad tails that are efficiently rejected by the consistency cut.

Figure 7 shows the distribution of the selected bunch-offset index n_{best} for accepted events in the toy Monte Carlo (here for $\sigma_{\text{det}} = 3 \text{ ns}$ and $k = 5$). For each event, the algorithm evaluates the discrete set of candidate offsets from Eq. (45) and computes the corresponding time-of-flight residuals r_n (as defined above). The chosen offset n_{best} is defined as the candidate that yields the smallest absolute residual, i.e.

$$|r_{n_{\text{best}}}| = \min_{n \in \{0, \dots, n_{\text{max}}\}} |r_n|, \quad (49)$$

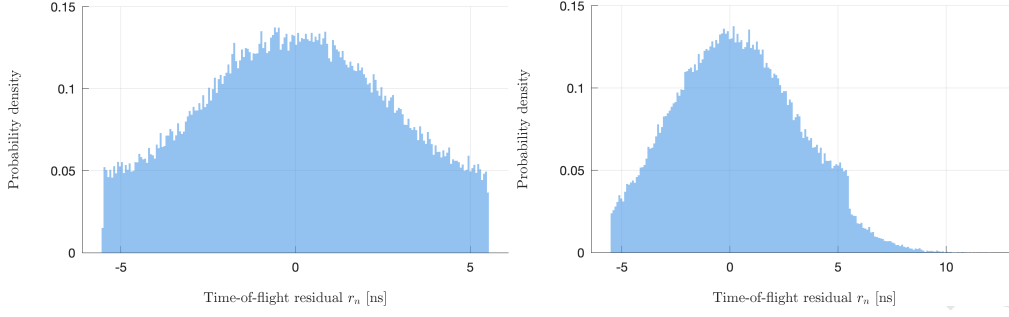


Figure 6: Normalized distributions of the time-of-flight residual r_n for accepted events at $|t| = 0.0010 \text{ (GeV/c)}^2$ (left) and $|t| = 0.0100 \text{ (GeV/c)}^2$ (right), obtained for a detector timing resolution of $\sigma_{\text{det}} = 3 \text{ ns}$. The vertical axis shows the probability density, such that the area under each histogram is unity.

and the event is accepted if this minimum residual falls inside the consistency window

$$\min_{n \in \{0, \dots, n_{\text{max}}\}} |r_n| < k \sigma_{\text{eff}}, \quad (50)$$

where k is a dimensionless scale factor that sets the half-width of the acceptance window in units of the effective timing resolution σ_{eff} . The distribution of n_{best} therefore quantifies how often nonzero bunch offsets contribute among accepted events and provides a direct check that the chosen search range n_{max} covers the relevant bunch hypotheses for the simulated kinematics.

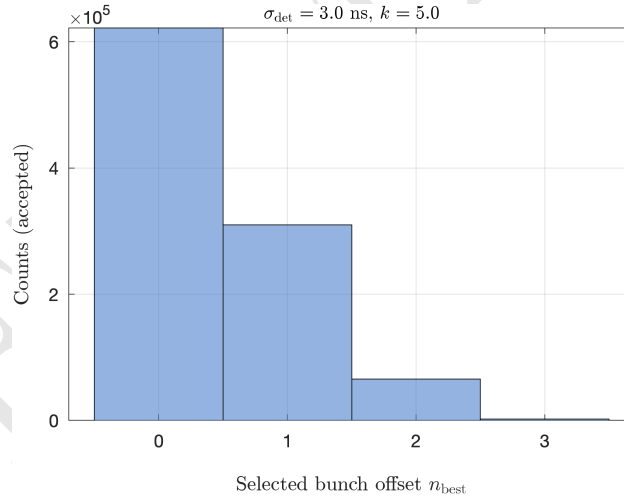


Figure 7: Distribution of the selected bunch-offset index n_{best} for accepted events in the toy Monte Carlo at $\sigma_{\text{det}} = 3 \text{ ns}$. Events are accepted when the minimum absolute time-of-flight residual satisfies $\min_n |r_n| < k \sigma_{\text{eff}}$ with $k = 5$, i.e., a $\pm 5 \sigma_{\text{eff}}$ consistency window around the predicted time of flight.

The impact of bunch mis-identification on the measured beam polarization can be estimated from dilution arguments. Each incorrectly assigned recoil event contributes to the measured asymmetry with random azimuthal angle, effectively diluting the true asymmetry signal. To first order, the measured polarization is related to the true beam polarization by

$$P_{\text{measured}} \approx (1 - f_{\text{wrong}}) P_{\text{true}}, \quad (51)$$

where f_{wrong} denotes the fraction of events with incorrect bunch assignment. This relation shows that the relative systematic uncertainty on the beam polarization measurement scales directly as

$$\frac{\Delta P}{P} \approx f_{\text{wrong}}. \quad (52)$$

The total systematic uncertainty budget for EIC polarimetry includes contributions from acceptance corrections, energy-scale calibration, and target polarization determination. To ensure that bunch mis-identification does not dominate the systematic uncertainty budget, a conservative requirement is $f_{\text{wrong}} < 0.001$ (0.1%), corresponding to $\Delta P/P < 0.1\%$ from this source alone. This allocation leaves adequate margin for other systematic contributions while meeting the overall EIC polarimetry goal of $\Delta P/P < 1\%$.

Figure 8 quantifies the systematic uncertainty from bunch mis-identification as a function of detector timing resolution. The results demonstrate that conventional silicon detector technology with timing resolutions of $\sigma_{\text{det}} \lesssim 1.5$ ns achieves bunch mis-identification fractions below 10^{-3} (0.1%), well within the systematic uncertainty budget for high-precision EIC polarimetry. For $\sigma_{\text{det}} < 1$ ns, the mis-identification fraction drops below 10^{-4} (0.01%), demonstrating that excellent performance is achievable with established detector technology. The plateau at excellent timing resolutions shows that the acceptance window $k = 5$ efficiently captures correctly identified events, while the steep transition around $\sigma_{\text{det}} \sim 1.5$ –2 ns demonstrates the effectiveness of the TOF-energy consistency criterion. This performance is achieved using a relatively wide acceptance window ($k = 5$, corresponding to $\pm 5\sigma_{\text{eff}}$), which ensures high efficiency while maintaining excellent purity in bunch assignment.

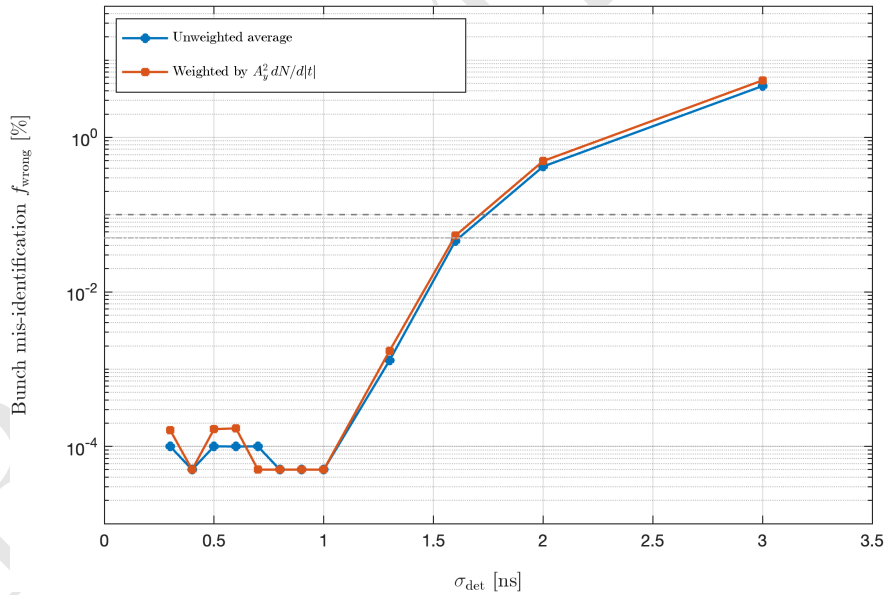


Figure 8: Bunch mis-identification fraction f_{wrong} as a function of detector timing resolution σ_{det} , evaluated with acceptance window $k = 5.0$. Both unweighted average (blue) and weighted average by $A_y^2 dN/d|t|$ (orange) are shown. Reference lines indicate the 0.1% and 0.05% systematic uncertainty targets. The steep transition around $\sigma_{\text{det}} \sim 1$ –2 ns demonstrates that sub-nanosecond timing is not required to achieve sub-permille level systematic uncertainties in bunch assignment.

Figure 8 shows both the unweighted average of f_{wrong} across all $|t|$ bins and a physics-weighted average that accounts for the expected event distribution and analyzing power in the CNI region [2].

The close agreement between the weighted and unweighted averages across the full range of detector timing resolutions demonstrates that bunch mis-identification performance is relatively uniform across the CNI kinematic region. The weighted average shows slightly higher f_{wrong} values at poor timing resolution ($\sigma_{\text{det}} > 2 \text{ ns}$), reflecting the fact that low- $|t|$ events (which have longer time-of-flight and are therefore more susceptible to mis-identification) receive significant weight in the physics analysis due to their large analyzing power and event yield.

5.7.2 Achievable timing resolution with waveform-digitizer readout

This subsection documents the technical basis for the detector timing resolution assumed in the toy Monte Carlo studies of recoil–bunch assignment performance. As demonstrated in Fig. 8, detector timing resolutions of $\sigma_{\text{det}} \lesssim 1.5 \text{ ns}$ are sufficient to achieve bunch mis-identification systematics below the 0.1% level. This subsection demonstrates that such performance is technically realistic and achievable using modern silicon detectors in combination with waveform-digitizer-based readout.

A key element of precise timing extraction from digitized detector signals is constant-fraction discrimination (CFD). In a conventional analog CFD, the signal time is defined at the point where the pulse reaches a fixed fraction of its maximum amplitude, rather than at a fixed voltage threshold, thereby suppressing time walk associated with pulse-height variations. The same principle can be implemented digitally (digital CFD, or dCFD) by operating directly on sampled waveforms: a digitally transformed signal is constructed from delayed and scaled samples, and the zero crossing of this transformed waveform defines the hit time. Interpolation between adjacent samples around the zero crossing yields timing information with sub-sample precision. Detailed descriptions of dCFD algorithms and their performance in waveform digitizers are given in Ref. [15] and references therein.

The intrinsic timing scale set by the silicon sensor is determined primarily by charge-collection dynamics and the resulting signal rise time. For a fully depleted silicon detector of thickness d , the characteristic charge-collection time can be estimated as

$$t_{\text{coll}} \approx \frac{d}{v_d}, \quad (53)$$

where v_d denotes the carrier drift velocity. At typical operating bias voltages of order 150 V to 300 V for silicon sensors of this thickness, corresponding to electric fields of several kV/cm, the electron drift velocity in silicon approaches saturation at $v_{e,\text{sat}} \sim 10^7 \text{ cm/s}$, which governs the early leading-edge timing relevant for hit-time extraction. For a detector thickness of $d \simeq 320 \mu\text{m}$, representative of the Hamamatsu S13804 series single-sided silicon strip detectors foreseen for the recoil system [13, 14], this yields a characteristic charge-collection time of order $t_{\text{coll}} \sim 3 \text{ ns}$. This time scale defines the natural rise time of the detector signal but does not by itself set the achievable timing resolution.

The achievable timing resolution is instead dominated by the readout chain and the time-extraction algorithm. Modern waveform digitizers sampling at rates of approximately 250 MSa/s to 1000 MSa/s provide sufficient time granularity to resolve the leading edge of silicon detector signals. In this regime, the timing resolution is not limited by the sampling interval itself, but by the signal-to-noise ratio, the effective bandwidth of the front-end electronics, and the stability of the timing algorithm. Established techniques such as dCFD, template fitting, or leading-edge fits applied to digitized waveforms routinely achieve timing resolutions at or below the nanosecond level for detector signals with comparable rise times and bandwidth [15, 16].

Additional contributions from clock distribution jitter and intrinsic digitizer timing uncertainty are typically at the level of a few hundred picoseconds for modern systems and are therefore

subdominant in the present application. Taken together, these considerations demonstrate that an overall detector timing resolution of order 2 ns, as assumed in the toy Monte Carlo studies of recoil-bunch assignment, represents a conservative and technically well-justified performance target rather than an aggressive requirement.

The toy Monte Carlo results presented in Fig. 8 establish that nanosecond-scale timing resolution is sufficient to achieve negligible systematic uncertainties from bunch mis-identification. Detailed studies of recoil transport, acceptance corrections, and systematic uncertainties using full detector simulations will be addressed in subsequent work.

Taken together, the results presented here demonstrate that the proposed recoil detector concept satisfies the physics, timing, and operational requirements for high-precision polarization measurements in the EIC CNI region.

5.8 Detector concept readiness and path forward

The detector concept presented in this section has reached a level of technical maturity appropriate for a Preliminary Design Review. The results reported here establish a consistent baseline description of recoil detection, timing performance, and recoil–bunch association for the proposed HJET detector system under EIC operating conditions. The treatment intentionally focuses on the dominant kinematic and timing effects governing bunch assignment performance and detector feasibility in the Coulomb–nuclear interference region.

The overall detector layout and sensor technology are based on established solutions. Silicon strip detectors with thicknesses of order 300 μm provide adequate energy resolution, segmentation, and rate capability for recoil proton detection, while remaining compatible with operation in magnetic holding fields of several hundred millitesla. The compact detector geometry at a radial distance of approximately 220 mm significantly increases the effective azimuthal acceptance compared to the present RHIC HJET system, thereby improving statistical precision without introducing new acceptance-driven systematic effects.

Timing requirements for recoil-bunch association have been derived from the EIC bunch structure and recoil kinematics and validated using a dedicated toy Monte Carlo simulation. These studies demonstrate that robust bunch identification with systematic uncertainties below 0.1% can be achieved over the full CNI kinematic range for detector timing resolutions $\sigma_{\text{det}} \lesssim 1.5$ ns, as quantified in Fig. 8.

Several additional effects that are known to be present in a realistic detector environment have been deliberately omitted at this stage in order to keep the performance studies transparent and interpretable. These effects are not expected to qualitatively alter the conclusions drawn here, but may influence quantitative performance estimates and will be incorporated in future refinements of the simulation framework. Relevant effects include recoil transport in the magnetic holding field and the associated path-length variations, detailed energy-loss straggling and angular multiple scattering in the sensor material, realistic detector segmentation and clustering effects, correlations introduced by full waveform digitization and signal processing, as well as detector alignment, magnetic-field non-uniformities, and geometry-dependent acceptance effects.

For completeness, the following effects have been identified as natural extensions of the present toy Monte Carlo framework:

- **Magnetic-field-dependent recoil transport**, including trajectory curvature, effective path-length variations, and the impact of non-uniform holding fields.
- **Tracking-based path-length reconstruction**, enabling event-by-event determination of L_{path} rather than the use of a fixed effective flight path.

- **Energy smearing and detector-response correlations**, and their combined impact on the time-of-flight-energy consistency condition.
- **Full detector geometry and acceptance effects**, such as edge effects, azimuthal segmentation, and inactive material.
- **Backgrounds and pileup**, including non-CNI recoils, accidental hits, and electronic noise.

At the level of this study, these effects are expected to modify quantitative performance metrics but not the qualitative conclusions regarding the viability of recoil-bunch assignment based on time-of-flight and recoil-energy consistency.

No fundamental technical risks have been identified that would preclude implementation of the proposed detector concept. The remaining work toward a Final Design Review is therefore primarily concerned with systematic refinement, engineering optimization, and experimental validation rather than conceptual development. Key next steps include laboratory measurements of timing performance using representative silicon sensors and waveform digitizers, validation of detector operation and signal integrity in magnetic holding fields comparable to the design field strength, and detailed integration studies of front-end electronics, clock distribution, and data acquisition within the HJET infrastructure.

Taken together, the results presented in this chapter demonstrate that the proposed detector system is technically sound, internally consistent, and compatible with the operational constraints of the EIC HJET polarimeter. The detector concept therefore provides a solid and credible basis for advancing to detailed engineering design and targeted prototyping in preparation for a Final Design Review.

6 Summary

This document presents the conceptual design and performance studies of a compact, switchable three axis magnetic holding field system for the HJET polarimeter at the Electron-Ion Collider. The primary motivation is to suppress beam induced depolarization of the atomic hydrogen target by increasing the holding field strength from the RHIC baseline of 120 mT to values of up to 400 mT, while preserving the recoil proton detector acceptance required for Coulomb-Nuclear Interference polarimetry. This upgrade is required to enable high precision absolute beam polarimetry at the EIC with a target precision of $\Delta P/P \approx 1\%$ based on reliable target polarization control and polarity reversal.

The baseline configuration employs three orthogonal pairs of superconducting coils to generate the field components $\pm B_x$, $\pm B_y$, and $\pm B_z$, with only one component energized at a time. The transverse coil pairs are positioned at $x, y = \pm 80$ mm, while the longitudinal holding field is generated by a separate upstream downstream coil pair. This compact geometry maintains a material-free central region of roughly 100 mm diameter around the ion beam axis and is compatible with the required detector layout and acceptance constraints.

Three dimensional Biot-Savart calculations using realistic coil geometries demonstrate that a holding field of 400 mT can be achieved in the target region with sufficient uniformity using 100 turns per coil at operating currents of approximately 1.6 kA. The corresponding stored magnetic energy, of order 5 kJ to 6 kJ per coil pair, simplifies quench protection and supports rapid field ramping and axis switching.

Beam and spin dynamics constraints associated with the magnetic holding field system have been identified but are not treated in detail in the present work. In particular, the requirement

of a vanishing integrated magnetic field along the circulating ion beam orbit, $\int B \cdot d\ell \approx 0$, may necessitate additional upstream and downstream compensation elements. Furthermore, spin rotations induced by the holding field as described by the Thomas-BMT formalism and transverse orbit kicks arising from residual B_x and B_y field components must be evaluated against machine optics constraints. These effects define boundary conditions for the final integrated system design and will be addressed at a later stage.

A quantitative comparison with the present RHIC HJET configuration is derived from the effective sensitivity for transverse polarization measurements, taking into account the finite detector acceptance and the azimuthal weighting relevant for the extraction of transverse polarization components. For the proposed full EIC system, six detector modules of the same active size are placed at a radius of $r \approx 220$ mm. The resulting effective gain corresponds to a reduction of the measurement time at fixed statistical precision by a factor of about 6 to 7 relative to the RHIC HJET configuration, which employs two detectors of the same active size located at $r \approx 750$ mm. In addition, the expected increase of the EIC proton beam intensity by approximately a factor of three provides a further reduction of the required measurement time. Taken together, an overall reduction of the measurement time by approximately a factor of $6.8 \times 3 \approx 20$ can therefore be expected relative to the present RHIC HJET configuration for the same statistical precision.

The impact of the guide field on recoil proton transport has been evaluated using numerical trajectory calculations. The resulting transverse displacements at the detector locations remain compatible with the proposed acceptance and detector size, confirming that the magnetic field required for target polarization control does not compromise recoil detection in the CNI region.

A dedicated Monte Carlo study has been developed to assess recoil bunch assignment based on the combined consistency of time of flight and recoil energy under realistic EIC timing conditions. The study incorporates detector level timing assumptions appropriate for waveform digitizer based readout of silicon strip detectors, including finite charge collection times, signal rise times, and timing extraction from sampled waveforms using established techniques such as digital constant fraction discrimination. On this basis, detector timing resolutions at the level of a few nanoseconds are found to be both technically realistic and conservative.

The resulting bunch assignment performance has been evaluated over the full Coulomb-Nuclear Interference kinematic range, demonstrating high correct assignment fractions even in the presence of the EIC bunch structure. These results confirm that precise recoil identification and unambiguous bunch association can be achieved without imposing aggressive or unrealistic requirements on the detector readout or clock distribution, supporting the overall feasibility of the proposed polarimetry scheme.

The system is designed for staged deployment to allow early operation and incremental expansion. An initial Phase 1 implementation provides a single transverse guide field component using one coil pair and a single bipolar power supply, enabling commissioning and physics operation with reduced hardware scope. Phase 2 extends the system to full three axis capability through the addition of the remaining coil pairs. Depending on the choice of superconducting technology and final engineering implementation, the projected cost envelope spans approximately 150 k\$ to 280 k\$ for Phase 1 and 330 k\$ to 650 k\$ for the complete three axis system.

The remaining work required to advance the proposed system beyond the present stage is well defined and limited in scope. It includes detailed mechanical tolerance and alignment studies for the coil and detector supports, a systematic evaluation of magnetic field uniformity requirements and their correlation with polarization related systematic uncertainties, and the integration of the full detector geometry and readout response into higher fidelity simulations. In addition, engineering studies addressing coldhead based cooling integration, thermal interfaces, power supply design and qualification, quench protection, and installation constraints will be required as part of the

subsequent design phase.

At the level presented here, the magnetic field concept, detector integration, recoil transport studies, timing requirements, and cost envelope are defined with sufficient detail and internal consistency to support a Preliminary Design Review. The results demonstrate technical feasibility, controlled performance margins, and a clear path toward an engineering ready implementation of the proposed guide field system.

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A Magnetic field of a square current loop

We derive the magnetic field along the axis of a square current loop using the Biot–Savart law in Eq. (6). Consider a square loop of side length L lying in the xy plane centered at the origin, with corners at $(\pm L/2, \pm L/2, 0)$. The loop carries current I in the counterclockwise direction when viewed from positive z .

A.1 Biot-Savart law

The magnetic field at position $\mathbf{r}$ due to a current element $I d\ell'$ at position $\mathbf{r}'$ is

$$d\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \frac{d\ell' \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}. \quad (54)$$

For a field point on the z axis at $(0, 0, z)$, by symmetry, only the z component of the field survives. We compute the contribution from each of the four sides.

A.2 Side 1: Bottom edge parallel to x axis

This side runs from $(-L/2, -L/2, 0)$ to $(+L/2, -L/2, 0)$ with current in the $+x$ direction.

Position on wire: $\mathbf{r}' = (x, -L/2, 0)$ where $-L/2 \leq x \leq L/2$

Current element: $d\ell' = dx \vec{e}_x$

Vector to field point: $\mathbf{r} - \mathbf{r}' = (0, 0, z) - (x, -L/2, 0) = (-x, L/2, z)$

Distance: $|\mathbf{r} - \mathbf{r}'| = \sqrt{x^2 + (L/2)^2 + z^2}$

1245 Cross product:

$$d\ell' \times (\mathbf{r} - \mathbf{r}') = dx \vec{e}_x \times \left(-x \vec{e}_x + \frac{L}{2} \vec{e}_y + z \vec{e}_z \right) = dx \left(-z \vec{e}_y + \frac{L}{2} \vec{e}_z \right). \quad (55)$$

1246 The z component contribution is

$$dB_z^{(1)} = \frac{\mu_0 I}{4\pi} \frac{(L/2) dx}{[x^2 + (L/2)^2 + z^2]^{3/2}}. \quad (56)$$

1247 Integrating from $x = -L/2$ to $x = +L/2$ yields

$$B_z^{(1)} = \frac{\mu_0 IL}{8\pi} \int_{-L/2}^{+L/2} \frac{dx}{[x^2 + (L/2)^2 + z^2]^{3/2}}. \quad (57)$$

1248 Using the standard integral

$$\int \frac{dx}{(x^2 + a^2)^{3/2}} = \frac{x}{a^2 \sqrt{x^2 + a^2}}, \quad (58)$$

1249 with $a^2 = (L/2)^2 + z^2$, we obtain

$$B_z^{(1)} = \frac{\mu_0 IL}{8\pi} \left[\frac{x}{a^2 \sqrt{x^2 + a^2}} \right]_{-L/2}^{+L/2}. \quad (59)$$

1250 Since $\sqrt{x^2 + a^2}$ is even in x , the bracket evaluates to

$$\left[\frac{x}{a^2 \sqrt{x^2 + a^2}} \right]_{-L/2}^{+L/2} = \frac{L}{a^2 \sqrt{(L/2)^2 + a^2}}, \quad (60)$$

1251 and therefore

$$B_z^{(1)} = \frac{\mu_0 IL^2}{8\pi a^2 \sqrt{(L/2)^2 + a^2}} = \frac{\mu_0 IL^2}{8\pi [(L/2)^2 + z^2] \sqrt{z^2 + L^2/2}}. \quad (61)$$

1252 A.3 Contributions from all four sides

1253 By symmetry, each of the four sides contributes equally to B_z . The total field is

$$B_z(z) = 4B_z^{(1)} = \frac{\mu_0 IL^2}{2\pi [(L/2)^2 + z^2] \sqrt{z^2 + L^2/2}}. \quad (62)$$

1254 Introducing $u = 2z/L$ and factoring out L , this can be written as

$$B_z(z) = \frac{4\mu_0 I}{\pi L} \frac{1}{(1 + u^2) \sqrt{u^2 + 2}} = \frac{4\mu_0 I}{\pi L} \frac{1}{[1 + (2z/L)^2] \sqrt{2 + (2z/L)^2}}. \quad (63)$$

1255 At the center ($z = 0$), we get

$$B_z(0) = \frac{4\mu_0 I}{\pi L} \frac{1}{\sqrt{2}} = \frac{2\sqrt{2} \mu_0 I}{\pi L}. \quad (64)$$

1256 This completes the derivation of Eqs. (7) and (8).

B Estimate of hit displacements on the detector surface

We quantify the magnetic deflection of recoil protons traversing the field of a circular coil pair in a Helmholtz-like configuration. The calculation combines a numerical field evaluation with a momentum-kick integration of the proton trajectory and is benchmarked against a uniform reference field.

B.1 Magnet geometry and field

Two identical circular coils of radius

$$R = 33.1 \text{ mm}, \quad (65)$$

lie in the xz plane, with their centers at $y = \pm 80 \text{ mm}$. The coil axis is along y . The target holding field at the origin is

$$B_y(0, 0, 0) = +0.4 \text{ T}. \quad (66)$$

For a circular loop on its symmetry axis, the magnetic field is

$$B_y(0) = \frac{\mu_0}{2} \frac{IR^2}{(R^2 + y_0^2)^{3/2}}. \quad (67)$$

For two identical coils this becomes

$$B_y^{\text{pair}}(0) = \mu_0 I \frac{R^2}{(R^2 + y_0^2)^{3/2}}. \quad (68)$$

For the present geometry, this corresponds to a required current of

$$I = 1.89 \times 10^5 \text{ A}, \quad (69)$$

for a single-turn coil pair. The off-axis field $B_y(x, 0, 0)$ is evaluated numerically using a discretized Biot-Savart integration over the circular loops. The current direction is chosen such that $B_y(0, 0, 0) > 0$, fixing the sign convention unambiguously.

B.2 Recoil proton kinematics

The recoil proton is assumed to be initially at rest. For elastic scattering at small momentum transfer t in the CNI regime, the recoil kinetic energy is

$$T_R = \frac{|t|}{2m_p}, \quad (70)$$

with $m_p = 0.938 \text{ GeV}/c^2$. The exact relativistic recoil momentum is

$$p_R = \sqrt{T_R(T_R + 2m_p)} \simeq \sqrt{|t|}, \quad (71)$$

where the approximation is valid over the t range considered here.

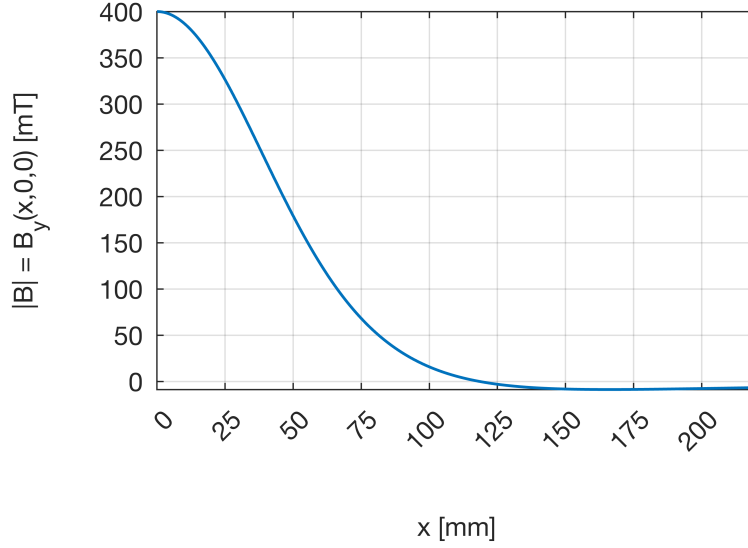


Figure 9: Magnetic field on the symmetry line $y = 0$, $z = 0$, shown as $B_y(x, 0, 0)$ for a circular coil pair with $R = 33.1$ mm and coil centers at $y = \pm 80$ mm. By symmetry the magnetic field on this line is purely directed along y , so that B_y equals the total field magnitude.

B.3 Trajectory integration

The proton motion is predominantly along the $+x$ direction. With $\mathbf{B} = B_y \vec{e}_y$, the Lorentz force produces a transverse momentum kick in z ,

$$dp_z = -q B_y(x) dx. \quad (72)$$

The local trajectory slope is $\theta(x) = p_z/p$, and the transverse displacement is updated according to

$$z_{i+1} = z_i + \theta_i dx. \quad (73)$$

This momentum-kick formulation is dimensionally transparent and avoids explicit use of v or γ . The magnetic field is evaluated either on the $z = 0$ axis or self-consistently along the actual trajectory $B_y(x, 0, z)$; for the present parameters both approaches give comparable results.

B.4 Sanity check with a uniform box field

As a unit and scale check, consider a uniform magnetic field

$$B = 0.4 \text{ T} \quad (74)$$

acting over a length

$$L = 50 \text{ mm}, \quad (75)$$

followed by a drift to a detector at $x = 250$ mm. The resulting transverse deflection is

$$z \approx \frac{qBL}{p} \left(x_{\text{det}} - \frac{L}{2} \right). \quad (76)$$

This estimate yields deflections of the same order as the full coil-field calculation.

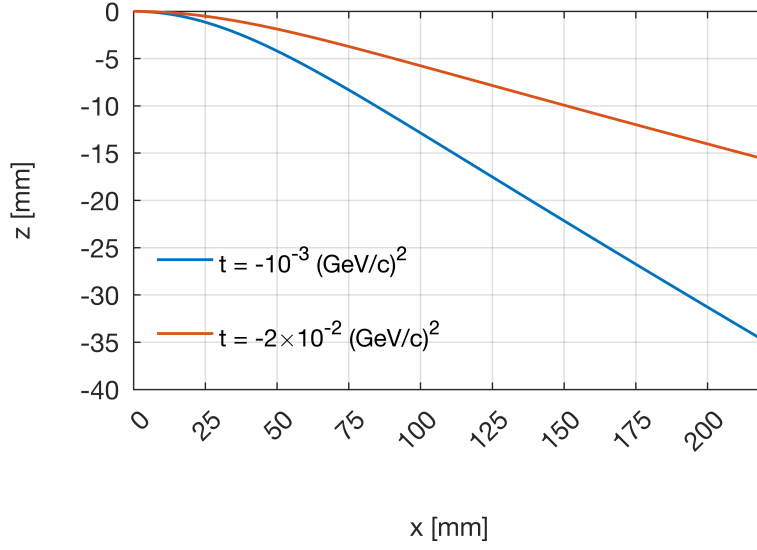


Figure 10: Transverse displacement $z(x)$ of recoil protons for two representative values of $|t|$ calculated in the magnetic field generated by the circular coil pair with $R = 33.1$ mm and $y_0 = 80$ mm with the detector place at $x = 220$ mm.

For the present parameters, the box-field estimate yields explicit numerical values. For a uniform field $B = 0.4$ T acting over $L = 50$ mm and a detector position at $x = 220$ mm, the expected transverse displacements are $|\Delta z| \approx 37.0$ mm for $|t| = 10^{-3} (\text{GeV}/c)^2$ and $|\Delta z| \approx 16.5$ mm for $|t| = 5 \times 10^{-3} (\text{GeV}/c)^2$. The full coil-field calculation in Fig. 10 yields corresponding values of $\Delta z \approx -34.9$ mm and $\Delta z \approx -15.6$ mm, respectively.